



The Infinite Bandwidth Company™

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Micrel

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Corporate Profile

Micrel Semiconductor is a leading manufacturer of advanced, high performance communications, clock management, mixed signal, analog and power ICs. These products include devices for fiber-optic telecommunications and networking, cellular telephones, servers, portable computer systems, computer peripherals, process control systems, consumer electronics and power supplies.

Founded in 1978, Micrel has a portfolio of world-class wafer fabrication processes that provide the ability to produce new products faster than our competition. We use singularly or in combination CMOS, Bipolar and CMOS technologies in our analog products. With the acquisition of Synergy Semiconductor in 1998, we now have high-speed bipolar capability in the form of our ASSET (All Spacer Separated Element Transistor) process. This proprietary, patented bipolar technology employs a narrow deep-trench isolation technique allowing for electrical isolation between adjacent circuit elements. This means we can put multiple analog and digital functions on the same IC without the various functions causing interference found in alternate techniques.

Communication

Our family of communications ICs is comprised of two primary components. The post amps and laser drivers designed for fibre optic modules are used in many Gigabit Ethernet, Fibre Channel and GBIC modules. The growing family of physical layer devices is compliant to the demanding SONET/SDH jitter specifications.

Logic and Clocks

Micrel offers an alternate source for ON Semiconductor's industry standard ECLinPS™ and ECLinPS Lite™ ECL logic families. In many cases we provide performance advantages in the form of higher speed. Our ClockWorks™ family offers a complete toolkit to allow the construction of ultra-low jitter clock systems.

Low-Dropout Voltage Regulators

Micrel has gained a reputation as a leading vendor of LDO regulators and is a major supplier to the mobile phone and computing industries. Micrel has a very broad portfolio of LDO regulators ranging from 80mA devices in IttyBitty® SOT23-5 packages to 7.5A power devices and LDO controllers for even higher current applications.

Radio Frequency Products

Our family of QwikRadio™ RF Receivers provides users with the industry's most highly-integrated, easy-to-use way of adding low-data-rate RF communications to a system. The tiny SO-8 devices require just three external components to build an entire superheterodyne receiver.

Universal Serial Bus and PCMCIA

The recognized leader in providing USB and PC Card power distribution solutions, Micrel's extensive portfolio of products are already in use by many of the leading PC and peripheral OEMs.

MOSFET Drivers

Micrel offers a broad range of MOSFET drivers, ranging from IttyBitty® devices in SOT-23 packages to 12A high-power devices.

Switch-Mode Power Supply Products

Micrel has a rapidly expanding portfolio of switching regulator products, servicing computer, telecommunications and industrial accounts. All products combine Micrel's advanced processes with strong design talent to produce best-in-class products.

Micrel addresses step-up, step-down and multioutput systems, all offering speed and efficiency advantages.

Custom and Semicustom Products

More than just a standard products company, Micrel has strong custom and semicustom capabilities. The innovative MPD8021 EZAnalog® provides all the design tools over the Internet for users defining their own custom analog IC.

Other Product Lines

- Operational amplifiers and comparators
- Charge pumps
- Voltage references
- Miniature MOSFETs
- Serial and parallel-input latched drivers

ISO9001 Quality Assurance

Micrel first achieved ISO 9001 registration on March 31, 1997 and has maintained that registration through surveillance audits. We are committed to delivering products and services that meet or exceed our customers' expectations—error free, on-schedule, and at a competitive price. At Micrel we have a culture of continuous improvement that runs through all disciplines in the company and includes the activities of our suppliers.

Commitment to Customer Satisfaction

We are committed to helping our customers be successful. Our internal processes have been developed with flexibility in mind, so that we can quickly react to your changing requirements. Our outstanding sales, customer service and technical support organizations make it easy to get your questions answered.

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LDO Regulator Selection Guide

SC70 Package, $\theta_{JA} = 450^{\circ}\text{C/W}$

Device	I _{OUT}	V _{OUT}	V _{IN}	I _{GND} (typ.)	V _{DROPOUT} (typ.)	Comments
MIC5213	80mA	2.5, 2.8, 3.0, 3.3, 3.6	2.5V to 16V	180µA	300mV	µCap LDO
SOT23-5 Package, $\theta_{JA} = 235^{\circ}\text{C/W}$						
MIC5231	10mA	2.75, 3.0, 3.3, 5.0	3.5V to 12V	650nA	150mV	µCap LDO
MIC5203	80mA	2.8, 3.0, 3.3, 3.6, 3.8, 4.0, 4.5, 4.75, 5.0	2.5V to 16V	180µA	300mV	µCap LDO
MIC5270	100mA	–3.0, –4.1, –5.0, Adj.	–2V to –16V	70µA	480mV	Negative µCap LDO
MIC5205	150mA	2.7, 2.8, 3.0, 3.1, 3.2, 3.3, 3.6, 3.8, 4.0, 5.0, Adj.	2.5V to 16V	80µA	165mV	Low Noise LDO
MIC5206	150mA	2.5, 2.7, 3.0, 3.2, 3.3, 3.6, 3.8, 4.0, 5.0	2.5V to 16V	80µA	165mV	Low Noise LDO w/Error Flag
MIC5245	150mA	2.5, 2.6, 2.7, 2.8, 2.85, 3.0, 3.1, 3.2, 3.3	2.7V to 6V	100µA	150mV	Low Noise µCap LDO
MIC5246	150mA	2.6, 2.7, 2.8, 2.85, 3.0, 3.3	2.7V to 6V	85µA	150mV	Low Noise µCap LDO
MIC5247	150mA	1.5, 1.8, 2.0, 2.4	2.7V to 6V	85µA	150mV	µCap LDO
MIC5207	180mA	1.8, 2.5, 3.0, 3.3, 3.6, 3.8, 4.0, 5.0, Adj.	2.5V to 16V	80µA	165mV	Low Noise LDO
MIC5216	500mA	3.0, 3.3, 3.6, 5.0	2.5V to 12V	80µA	300mV	Low Noise LDO w/Error Flag
MIC5219	500mA	2.5, 2.7, 2.8, 3.0, 3.3, 3.6, 5.0, Adj.	2.5V to 12V	80µA	300mV	Low Noise LDO

MSOP-8 Package, $\theta_{JA} = 200^{\circ}\text{C/W}$

MIC5200	100mA	3.3, 5.0	2.5V to 26V	130µA	230mV	
MIC2951	150mA	5.0, Adj.	2V to 30V	120µA	300mV	Load Dump Protection
MIC5206	150mA	3.0, 3.3, 3.6, 3.8, 4.0, 5.0, Adj.	2.5V to 16V	80µA	165mV	Low Noise LDO w/Error Flag

T0-92 Package, $\theta_{JA} = 180^{\circ}\text{C/W}$

LP2950	100mA	5.0, 5.0 (0.5%)	2.0V to 30V	100µA	380mV	2nd Source to LP2950
MIC2950	150mA	5.0, 5.0 (0.5%)	2.0V to 30V	120µA	300mV	Load Dump Protection
MIC5207	180mA	3.3	2.5V to 16V	80µA	165mV	Low Noise LDO
MIC2954	250mA	5.0, 5.0 (0.5%)	2.0V to 30V	140µA	375mV	Load Dump Protection

SO-8 Package, $\theta_{JA} = 160^{\circ}\text{C/W}$

LP2951	100mA	4.85, 5.0, 5.0 (0.5%)	2.0V to 30V	100µA	380mV	2nd Source to LP2951
MIC5200	100mA	3.0, 3.3, 4.85, 5.0	2.5V to 26V	130µA	230mV	
MIC2951	150mA	3.3, 4.85, 5.0, 5.0 (0.5%)	2.0V to 30V	120µA	300mV	Load Dump Protection
MIC5201	200mA	3.0, 3.3, 5.0	2.5V to 26V	130µA	270mV	Load Dump Protection
MIC5201	200mA	Adj.	2.5V to 26V	130µA	270mV	Load Dump Protection
MIC2954	250mA	5.0, 5.0 (0.5%) Adj.	2.0V to 30V	140µA	375mV	Load Dump Protection

Device	I _{OUT}	V _{OUT}	V _{IN}	I _{QND} (typ.)	V _{DROPOUT} (typ.)	Comments
SO-8 Package, $\theta_{JA} = 160^{\circ}\text{C/W}$						
MIC29201	400mA	3.3 and Adj.	4.3V to 26V	140µA	450mV	Load Dump Protection
MIC29204	400mA	5.0 and Adj.	4.3V to 26V	140µA	450mV	Load Dump Protection
PDIP Package, $\theta_{JA} = 130^{\circ}\text{C/W}$						
LP2951	100mA	5.0 (0.5%), 5.0 (1%), Adj.	2V to 30V	100µA	380mV	2nd Source to LP2951
MIC2951	150mA	3.3, 4.85, 5.0 (0.5%), 5.0 (1%)	2V to 30V	120µA	380mV	Load Dump Protection
MIC29204	400mA	5.0 and Adj.	4.3V to 26V	140µA	450mV	Load Dump Protection
Power MSOP-8 Package, $\theta_{JA} = 80^{\circ}\text{C/W}$ (2"CU); 130°C/W (Mln. Footprint)						
MIC5236	150mA	2.5, 3.0, 3.3, 5.0	2.3V to 30V	20µA	350mV	Load Dump Protected µCap LDO
MIC5216	500mA	3.0, 3.3, 3.6, 5.0	2.5V to 12V	80µA	300mV	Low Noise LDO w/Error Flag
MIC5219	500mA	3.0, 3.3, 3.6, 5.0, Adj.	2.5V to 12V	80µA	300mV	Low Noise LDO
Power SOP-8 Package, $\theta_{JA} = 63^{\circ}\text{C/W}$ (2"CU); 100°C/W (Mln. Footprint)						
MIC5245	150mA	3.3	2.7V to 6V	100µA	150mV	Low Noise µCap LDO
MIC5236	150mA	2.5, 3.0, 3.3, 5.0	2.3V to 30V	20µA	350mV	Load Dump Protected µCap LDO
MIC5209	500mA	1.8, 2.5, 3.0, 3.3, 5.0, Adj.	2.5V to 16V	80µA	300mV	Low Noise
MIC39101	1A	1.8, 2.5, 3.3, 5.0	2.25V to 16V	400µA	410mV	Ultra-Low Dropout
MIC39102	1A	Adj.	2.25V to 16V	400µA	410mV	Ultra-Low Dropout
SOT-223 Package, $\theta_{JA} = 50^{\circ}\text{C/W}$ (2"CU); 95°C/W (Mln. Footprint)						
MIC5200	100mA	3.0, 3.3, 4.85, 5.0	2.5V to 26V	130µA	230mV	
MIC5201	200mA	3.0, 3.3, 4.85, 5.0	2.5V to 26V	130µA	270mV	Load Dump Protection
MIC2954	250mA	5.0, 5.0 (0.5%)	2V to 30V	140µA	375mA	Load Dump Protection
MIC2920A	400mA	3.3, 4.85, 5.0, 12	4.3V to 26V	140µA	450mV	Load Dump Protection
MIC5209	500mA	2.5, 3.0, 3.3, 4.2, 5.0	2.5V to 16V	80µA	300mV	Low Noise LDO
MIC39100	1A	1.8, 2.5, 3.3, 5.0	2.25V to 16V	400µA	410mV	Ultra-Low Dropout
T0-220 Package, $\theta_{JA} = 55^{\circ}\text{C/W}$ (Free Air); $\theta_{JC} = 2^{\circ}\text{C/W}$						
MIC2954	250mA	5.0, 5.0 (0.5%)	2.0V to 30V	140µA	375mV	Load Dump Protection
MIC29201	400mA	3.3, 4.85, 5.0, 12	4.3V to 26V	140µA	450mV	Load Dump Protection
MIC29202	400mA	Adj.	4.3V to 26V	140µA	450mV	Load Dump Protection
MIC2920A	400mA	3.3, 4.85, 5.0, 12	4.3V to 26V	140µA	450mV	Load Dump Protection
MIC5237	500mA	2.5, 3.3, 5.0	2.5V to 16V	80µA	300mV	
MIC29371	750mA	3.3, 5.0, 12	4.3V to 26V	160µA	370mV	Load Dump Protection
MIC5237	500mA	2.5, 3.3, 5.0	2.5V to 16V	80µA	300mV	
MIC29371	750mA	3.3, 5.0, 12	4.3V to 26V	160µA	370mV	Load Dump Protection

Device	I _{OUT}	V _{OUT}	V _{IN}	I _{IND} (typ.)	V _{DROPOUT} (typ.)	Comments
T0-220 Package, $\theta_{JA} = 55^{\circ}\text{C/W}$ (Free Air); $\theta_{JC} = 2^{\circ}\text{C/W}$						
MIC29372	750mA	Adj.	4.3V to 26V	160 μ A	370mV	Load Dump Protection
MIC2937A	750mA	3.3, 5.0, 12	4.3V to 26V	160 μ A	370mV	Load Dump Protection
MIC2940A	1.25A	3.3, 5.0, 12	4.3V to 26V	240 μ A	400mV	Load Dump Protection
MIC2941A	1.25A	Adj.	4.3V to 26V	240 μ A	400mV	Load Dump Protection
MIC2915x	1.5A	3.3, 5.0, 12, Adj.	2.25V to 26V	225 μ A	350mV	Load Dump Protection
MIC39150/1	1.5A	1.8, 2.5	2.25V to 16V	300 μ A	373mV	Ultra-Low Dropout
MIC2930x	3A	3.3, 5.0, 12, Adj.	2.25V to 26V	225 μ A	370mV	Load Dump Protection
MIC39300/1	3A	1.8, 2.5	2.25V to 16V	5mA	350mV	Ultra-Low Dropout
MIC29310/2	3A	3.3, 5.0, Adj.	2.3V to 16V	400 μ A	600mV	
MIC29311	3A	5.1	2.3V to 16V	400 μ A	600mV	Load Dump Protection
MIC2950x	5A	3.3, 5.0, Adj.	2.25V to 26V	225 μ A	370mV	USB LDO
MIC29510/2	5A	3.3, 5.0, Adj.	2.3V to 16V	500 μ A	700mV	
MIC39500/1	5A	1.8, 2.5	2.25V to 16V	—	350mV	Ultra-Low Dropout
MIC2971x	7.5A	3.3, 5.0, Adj.	2.3V to 16V	1mA	700mV	
T0-263 Package, $\theta_{JA} = 25^{\circ}\text{C/W}$ (2" CU); $\theta_{JC} = 2^{\circ}\text{C/W}$						
MIC2954	250mA	5.0, 5.0 (0.5%)	2.0V to 30V	140 μ A	375mV	Load Dump Protection
MIC29201	400mA	3.3, 4.85, 5.0, 12	4.3 to 26V	140 μ A	450mV	Load Dump Protection
MIC29202	400mA	Adj.	4.3 to 26V	140 μ A	450mV	Load Dump Protection
MIC5237	500mA	2.5, 3.3, 5.0	2.5V to 16V	80 μ A	300mV	
MIC5209	500mA	1.8, 2.5, 3.0, 3.3, 5.0, Adj.	2.5V to 16V	80 μ A	300mV	Low Noise LDO
MIC29371	750mA	3.3, 5.0, 12	4.3V to 26V	160 μ A	370mV	Load Dump Protection
MIC29372	750mA	Adj.	4.3V to 26V	160 μ A	370mV	Load Dump Protection
MIC2937A	750mA	3.3, 5.0, 12	4.3V to 26V	160 μ A	370mV	Load Dump Protection
MIC2940A	1.25A	3.3, 5.0, 12	4.3V to 26V	240 μ A	400mV	Load Dump Protection
MIC2941A	1.25A	Adj.	4.3V to 26V	240 μ A	400mV	Load Dump Protection
MIC2915x	1.5A	3.3, 5.0, 12, Adj.	2.25V to 26V	225 μ A	350mV	Load Dump Protection
MIC39150/1	1.5A	1.8, 2.5	2.25V to 16V	300 μ A	375mV	Ultra-Low Dropout
MIC2930x	3A	3.3, 5.0, 12, Adj.	2.25V to 26V	225 μ A	350mV	Load Dump Protection
MIC39300/1	3A	1.8, 2.5	2.25V to 16V	5mA	350mV	Ultra-Low Dropout
MIC29310/2	3A	3.3, 5.0, Adj.	2.3V to 16V	400 μ A	600mV	
MIC29311	3A	5.1	2.3V to 16V	400 μ A	600mV	USB LDO
MIC2950x	5A	3.3, 5.0, Adj.	2.25V to 26V	225 μ A	370mV	Load Dump Protection
MIC39500/1	5A	1.8, 2.5	2.25V to 16V	—	350mV	Ultra-Low Dropout

Device	I _{OUT}	V _{OUT}	V _{IN}	I _{GND} (typ.)	V _{DROPOUT} (typ.)	Comments
T0-247 Package, $\theta_{JA} = 40^{\circ}\text{C/W}$ (Free Air); $\theta_{JC} = 1.5^{\circ}\text{C/W}$						
MIC2975x	7.5A	3.3, 5.0, Adj.	2.5V to 26V	—	425mV	Load Dump Protection
Dual LDOs						
SOT23-6 Package, $\theta_{JA} = 235^{\circ}\text{C/W}$						
MIC5211	50mA	1.8, 2.5, 2.7, 2.8, 3.3, 3.6, 5.0 1.8/2.5, 1.8/3.3, 2.5/3.3, 3.3/5.0	2.5V to 16V	90μA	250mV	μCap LDO
SO-8 Package, $\theta_{JA} = 160^{\circ}\text{C/W}$						
MIC5202	100mA	3.0, 3.3, 4.85, 5.0	2.5V to 26V	170μA	225mV	
MSOP-8 Package, $\theta_{JA} = 200^{\circ}\text{C/W}$						
MIC5208	50mA	3.0, 3.3, 3.6, 4.0, 5.0	2.5V to 16V	180μA	250mV	μCap LDO
MIC5210	150mA	2.7, 2.8, 3.0, 3.3, 3.6, 4.0, 5.0	2.5V to 16V	80μA	165mV	Low Noise LDO
MSOP-10 Package, $\theta_{JA} = 200^{\circ}\text{C/W}$						
MIC5250	150mA	2.7, 2.8, 3.0, 3.3	2.7V to 6V	100μA	250mV	Low Noise μCap LDO
LDO Controllers						
SOP-8 Package, $\theta_{JA} = 160^{\circ}\text{C/W}$						
MIC5156	—	3.3, 5.0, Adj.	3.0V to 36V	—	—	Drives External N-Channel MOSFET
SOP-14 Package, $\theta_{JA} = 120^{\circ}\text{C/W}$						
MIC5157	—	Selectable 3.3, 5.0, 12	3.0V to 36V	—	—	Drives External N-Channel MOSFET
MIC5158	—	5.0, 12	3.0V to 36V	—	—	Drives External N-Channel MOSFET

Switch-Mode Voltage Regulator Selection Guide

Device	Output Voltage Preferred				Topology	Input Voltage	Control Current	Frequency	Skip Mode	Shutdown	Sync	Overcurrent/ Current Limit	Thermal Protection	Package
	2.85	3.3	5.0	12 Adj.										
MIC2171	•				Boost	3V–40V	2.5A	100kHz		•			•	TO-220-5, TO-263-5
MIC2172	•				Boost	3V–40V	1.25A	100kHz			•	•	•	DIP, SOP-8
MIC3172	•				Boost	3V–40V	1.25A	100kHz			•	•	•	DIP, SOP-8
MIC2145	•				Boost	2.4V–16V	1.5A	450kHz			•	•	•	MSOP-8
MIC2186	•				Boost	2.9V–14V	External FET	400/200/100kHz	•	•	•	•	•	SOP-16
LM2574	•	•	•	•	Buck	4V–40V	0.5A	52kHz		•		•	•	DIP-8, SOP-14
MIC4574	•	•	•	•	Buck	4V–24V	0.5A	200kHz		•		•	•	DIP-8, SOP-14
LM2575	•	•	•	•	Buck	4V–40V	1A	52kHz		•		•	•	TO-220-5, TO-263-5, DIP-16, SOP-24
MIC4575	•	•	•	•	Buck	4V–24V	1A	200kHz		•		•	•	TO-220-5, TO-263-5
LM2576	•	•	•	•	Buck	4V–40V	3A	52kHz		•		•	•	TO-220-5, TO-263-5
MIC4576	•	•	•	•	Buck	4V–24V	3A	200kHz		•		•	•	TO-220-5, TO-263-5
MIC4680	•	•	•	•	Buck	4V–34V	1.3A	200kHz		•		•	•	SOP-8
MIC4681	•				Buck	4V–30V	2A	200kHz		•		•	•	SOP-8
MIC38C42	•				Isolated	15.5V–20V ⁽²⁾	External FET	500kHz				•		DIP-8/14, SOP-8/14, MSOP-8/14
MIC38C44	•				Isolated	9V–20V ⁽³⁾	External FET	500kHz				•		DIP-8/14, SOP-8/14, MSOP-8/14
MIC2570	(1)	(1)	(1)	(1)	•	Micropower Boost	1.3–15V	130mA	20kHz	•	•	•	•	SOP-8
MIC2571	(1)	(1)	(1)	(1)	•	Micropower Boost	0.9–15V	120mA	20kHz	•	•	•	•	MSOP-8
MIC2141	•				•	Micropower Boost	2.5V–14V	0.5A	330kHz		•	•	•	SOT23-5
MIC2142	•				•	Micropower Boost	2.2V–16V	0.5A	330kHz		•	•	•	SOT23-5
MIC2177	•	•	•	•	•	Synchronous Buck	4.5V–16.5V	2.5A	200kHz	•	•	•	•	SOP-20
MIC2178	•	•	•	•	•	Synchronous Buck	4.5V–16.5V	2.5A	200kHz	•	•	•	•	SOP-20
MIC2179	•	•	•	•	•	Synchronous Buck	4.5V–16.5V	1.5A	200kHz	•	•	•	•	SSOP-20
MIC2182	•	•	•	•	•	Synchronous Buck	4.5V–32V	External FETs	300kHz	•	•	•	•	narrow SOP-16, narrow SSOP-16
MIC2185	•				•	Synchronous Buck	2.9V–14V	External FET	400/200/100kHz	•	•	•	•	SOP-16

¹ 2.85V/3.3V/5V selectable

² 15.5V startup, 10V minimum operating

³ 9V startup, 8.2V minimum operating

PC Card / PCMCIA / CardBus Power Distribution Switch Selection Guide

Device	Slots Supported	V _{cc3} On-Resistance	V _{cc5} On-Resistance	V _{pp} On-Resistance	Current Limit	Thermal Shutdown	Charge* Pump	Package
MIC2557	Single			500mΩ				SOP-8
MIC2558	Dual			500mΩ				SOP-14
MIC2560	Single	40mΩ	70mΩ	550mΩ	■	■		SOP-16 Wide
MIC2561	Single	110mΩ	210mΩ	550mΩ	■	■		SOP-14
MIC2562A	Single	100mΩ	70mΩ	600mΩ	■	■	■	SOP-14
MIC2563A	Dual	100mΩ	70mΩ	600mΩ	■	■	■	SSOP-28
MIC2564A	Dual	120mΩ	85mΩ	1300mΩ	■	■	■	SSOP-24 TSSOP-24

* 12V not required for switching when equipped with a charge pump.

USB Transceiver Selection Guide

Device	Description	Min. Operating Voltage	Supported Speeds	Package
MIC2550	Universal Serial Bus Transceiver	2.5V	1.5Mb/s (low) and 12Mb/s (high)	TSSOP-14



USB and General Purpose Power Distribution Switch Selection Guide

Device	Type	Internal Charge Pump	Switch Element	Operating Voltage	Maximum Switch Current	Output Resistance	Body Diode Blocking	Open Load Detect	Enable Logic	Under-voltage Lockout	Current Limit	Thermal Shutdown	Fault Flag	Flag Transient Filter	Package
MIC2025-1	Single	■	N-channel	2.7V to 5.5V	500mA	140mΩ@5V	■	■	Noninverting	■	■	■	■	■	SOP-, MSOP-8
MIC2025-2	Single	■	N-channel	2.7V to 5.5V	500mA	140mΩ@5V	■	■	Inverting	■	■	■	■	■	SOP-, MSOP-8
MIC2026-1	Dual	■	N-channel	2.7V to 5.5V	500mA	140mΩ@5V	■	■	Noninverting	■	■	■	■	■	SOP-, DIP-8
MIC2026-2	Dual	■	N-channel	2.7V to 5.5V	500mA	140mΩ@5V	■	■	Inverting	■	■	■	■	■	SOP-, DIP-8
MIC2027-1	Quad	■	N-channel	2.7V to 5.5V	500mA	150mΩ@5V	■	■	Noninverting	■	■	■	■	■	SOP-16
MIC2027-2	Quad	■	N-channel	2.7V to 5.5V	500mA	150mΩ@5V	■	■	Inverting	■	■	■	■	■	SOP-16
MIC2075-1	Single	■	N-channel	2.7V to 5.5V	500mA	140mΩ@5V	■	■	Noninverting	■	■	Latched	■	■	SOP-, MSOP-8
MIC2075-2	Single	■	N-channel	2.7V to 5.5V	500mA	140mΩ@5V	■	■	Inverting	■	■	Latched	■	■	SOP-, MSOP-8
MIC2076-1	Dual	■	N-channel	2.7V to 5.5V	500mA	140mΩ@5V	■	■	Noninverting	■	■	Latched	■	■	SOP-, DIP-8
MIC2076-2	Dual	■	N-channel	2.7V to 5.5V	500mA	140mΩ@5V	■	■	Inverting	■	■	Latched	■	■	SOP-, DIP-8
MIC2077-1	Quad	■	N-channel	2.7V to 5.5V	500mA	150mΩ@5V	■	■	Noninverting	■	■	Latched	■	■	SOP-16
MIC2077-2	Quad	■	N-channel	2.7V to 5.5V	500mA	150mΩ@5V	■	■	Inverting	■	■	Latched	■	■	SOP-16
MIC2505	Single	■	N-channel	2.7V to 7.5V	2A	50mΩ@5V	■	■	Noninverting	■	■	■	■	■	SOP-8
MIC2505-1	Single	■	N-channel	2.7V to 7.5V	2A	50mΩ@5V	■	■	Noninverting	■	■	■	■	■	SOP-8
MIC2505-2	Single	■	N-channel	2.7V to 7.5V	2A	50mΩ@5V	■	■	Inverting	■	■	■	■	■	SOP-8
MIC2506	Dual	■	N-channel	2.7V to 7.5V	1A	125mΩ@5V	■	■	Noninverting	■	■	■	■	■	SOP-8
MIC2514	Single	■	P-channel	3V to 13.5V	400mA	2.4Ω@5V	■	■	Noninverting	■	■	■	■	■	SOT-23-5
MIC2524-1	Quad	■	N-channel	2.7V to 5.5V	500mA	140mΩ@5V	■	■	Noninverting	■	■	■	■	■	SOP-, DIP-16
MIC2524-2	Quad	■	N-channel	2.7V to 5.5V	500mA	140mΩ@5V	■	■	Inverting	■	■	■	■	■	SOP-, DIP-16
See MIC2027–MIC2524 not recommended for new designs.															
MIC2525-1	Single	■	N-channel	2.7V to 5.5V	500mA	140mΩ@5V	■	■	Noninverting	■	■	■	■	■	SOP-, DIP-8
MIC2525-2	Single	■	N-channel	2.7V to 5.5V	500mA	140mΩ@5V	■	■	Inverting	■	■	■	■	■	SOP-, DIP-8
See MIC2025–MIC2525 not recommended for new designs.															
MIC2526-1	Dual	■	N-channel	2.7V to 5.5V	500mA	140mΩ@5V	■	■	Noninverting	■	■	■	■	■	SOP-, DIP-8
MIC2526-2	Dual	■	N-channel	2.7V to 5.5V	500mA	140mΩ@5V	■	■	Inverting	■	■	■	■	■	SOP-, DIP-8
See MIC2026–MIC2526 not recommended for new designs.															
MIC2527-1	Quad	■	N-channel	2.7V to 5.5V	500mA	300mΩ@5V	■	■	Noninverting	■	■	■	■	■	SOP-, DIP-16
MIC2527-2	Quad	■	N-channel	2.7V to 5.5V	500mA	300mΩ@5V	■	■	Inverting	■	■	■	■	■	SOP-, DIP-16
See MIC2027–MIC2527 not recommended for new designs.															

Device	Type	Switch Element	Internal Charge Pump	Operating Voltage	Maximum		Body Diode Blocking	Open Load Detect	Enable Logic	Under -voltage		Current Limit	Thermal Shutdown	Fault Flag	Flag	
					Switch Current	Output Resistance				Lockout	Transient Filter				Package	
MIC2529-1	Single*	N-channel	■	2.7V to 5.5V	500mA	140mΩ@5V	■		Noninverting			■	■	■		SOP-8
MIC2529-2	Single*	N-channel	■	2.7V to 5.5V	500mA	140mΩ@5V	■		Inverting			■	■	■		SOP-8
MIC2536-1	Dual	N-channel	■	2.7V to 5.5V	100mA	700mΩ@5V	■		Noninverting			■	■	■	■	SOP-8, MSOP-8
MIC2536-2	Dual	N-channel	■	2.7V to 5.5V	100mA	700mΩ@5V	■		Inverting			■	■	■	■	SOP-8, MSOP-8
MIC2537-1	Quad	N-channel	■	2.7V to 5.5V	100mA	700mΩ@5V	■		Noninverting			■	■	■		SOP-16
MIC2537-2	Quad	N-channel	■	2.7V to 5.5V	100mA	700mΩ@5V	■		Inverting			■	■	■		SOP-16
MIC2544-1	Single	N-channel	■	2.7V to 5.5V	1.5A	120mΩ@5V	■		Noninverting			■	■	■		SOP-8,MSOP-8
MIC2544-2	Single	N-channel	■	2.7V to 5.5V	1.5A	120mΩ@5V	■		Inverting			■	■	■		SOP-8, MSOP-8
MIC2545A-1	Single	N-channel	■	2.7V to 5.5V	2.5A	50mΩ@5V	■		Noninverting			■	■	■		SOP-, DIP-8
MIC2545A-2	Single	N-channel	■	2.7V to 5.5V	2.5A	50mΩ@5V	■		Inverting			■	■	■		SOP-, DIP-8
MIC2546-1	Dual	N-channel	■	2.7V to 5.5V	1.5A	120mΩ@5V	■		Noninverting			■	■	■		SOP-16, TSSOP-16
MIC2546-2	Dual	N-channel	■	2.7V to 5.5V	1.5A	120mΩ@5V	■		Inverting			■	■	■		SOP-16, TSSOP-16
MIC2547-1	Dual	N-channel	■	2.7V to 5.5V	1.5A	120mΩ@5V	■		Noninverting			■	Latched	■		SOP-16, TSSOP-16
MIC2547-2	Dual	N-channel	■	2.7V to 5.5V	1.5A	120mΩ@5V	■		Inverting			■	Latched	■		SOP-16, TSSOP-16
MIC2548-1	Single	N-channel	■	2.7V to 5.5V	1.5A	120mΩ@5V	■		Noninverting			■	Latched	■		SOP-8, MSOP-8
MIC2548-2	Single	N-channel	■	2.7V to 5.5V	1.5A	120mΩ@5V	■		Inverting			■	Latched	■		SOP-8, MSOP-8
MIC2549A-1	Single	N-channel	■	2.7V to 5.5V	2.5A	50mΩ@5V	■		Noninverting			■	Latched	■		SOP-, DIP-8
MIC2549A-2	Single	N-channel	■	2.7V to 5.5V	2.5A	50mΩ@5V	■		Inverting			■	Latched	■		SOP-, DIP-8
* Single switch plus integrated 3.3V low-dropout linear regulator.																
USB Switches Supporting ACPI S0/S3 State Transitions**																
MIC2010-1	Dual	N-channel	■	4.5V to 5.5V	500mA	140mΩ@5V	■		Noninverting		■	■	■	■	■	QSOP-16
MIC2010-2	Dual	N-channel	■	4.5V to 5.5V	500mA	140mΩ@5V	■		Inverting		■	■	■	■	■	QSOP-16
MIC2012-1	Dual	N-channel	■	4.5V to 5.5V	500mA	140mΩ@5V	■		Noninverting		■	■	■	■	■	QSOP-16
MIC2012-2	Dual	N-channel	■	4.5V to 5.5V	500mA	140mΩ@5V	■		Inverting		■	■	■	■	■	QSOP-16
MIC2012	Dual	N-channel	■	4.5V to 5.5V	500mA	140mΩ@5V	■		—		■	■	■	■	■	SOP-8
MIC2070-1	Dual	N-channel	■	4.5V to 5.5V	500mA	140mΩ@5V	■		Noninverting		■	■	Latched	■	■	QSOP-16
MIC2070-2	Dual	N-channel	■	4.5V to 5.5V	500mA	140mΩ@5V	■		Inverting		■	■	Latched	■	■	QSOP-16
MIC2072-1	Dual	N-channel	■	4.5V to 5.5V	500mA	140mΩ@5V	■		Noninverting		■	■	Latched	■	■	QSOP-16
MIC2072-2	Dual	N-channel	■	4.5V to 5.5V	500mA	140mΩ@5V	■		Inverting		■	■	Latched	■	■	QSOP-16
MIC2072	Dual	N-channel	■	4.5V to 5.5V	500mA	140mΩ@5V	■		—		■	■	Latched	■	■	SOP-8

USB Switches Supporting ACPI S0/S3 State Transitions**

MIC2010-1	Dual	N-channel	■	4.5V to 5.5V	500mA	140mΩ@5V	■			Noninverting			■	■	■	■	■	QSOP-16
MIC2010-2	Dual	N-channel	■	4.5V to 5.5V	500mA	140mΩ@5V	■			Inverting			■	■	■	■	■	QSOP-16
MIC2012-1	Dual	N-channel	■	4.5V to 5.5V	500mA	140mΩ@5V	■			Noninverting			■	■	■	■	■	QSOP-16
MIC2012-2	Dual	N-channel	■	4.5V to 5.5V	500mA	140mΩ@5V	■			Inverting			■	■	■	■	■	QSOP-16
MIC2012	Dual	N-channel	■	4.5V to 5.5V	500mA	140mΩ@5V	■			—			■	■	■	■	■	SOP-8
MIC2070-1	Dual	N-channel	■	4.5V to 5.5V	500mA	140mΩ@5V	■			Noninverting			■	Latched	■	■	■	QSOP-16
MIC2070-2	Dual	N-channel	■	4.5V to 5.5V	500mA	140mΩ@5V	■			Inverting			■	Latched	■	■	■	QSOP-16
MIC2072-1	Dual	N-channel	■	4.5V to 5.5V	500mA	140mΩ@5V	■			Noninverting			■	Latched	■	■	■	QSOP-16
MIC2072-2	Dual	N-channel	■	4.5V to 5.5V	500mA	140mΩ@5V	■			Inverting			■	Latched	■	■	■	QSOP-16
MIC2072	Dual	N-channel	■	4.5V to 5.5V	500mA	140mΩ@5V	■			—			■	Latched	■	■	■	SOP-8

** Contact factory for availability of specific options. Specifications are given for "MAIN" mode operation.

Operational Amplifier Selection Guide

Device	Description	GBW	Slew Rate	Supply Current (per Op Amp)	Input Offset Voltage (max.)	Input Bias Current	Supply Range	Rail-to- Rail I/O	Package	Comments
MIC6211	High-Voltage Op Amp	2.5MHz	6V/μs	1.2mA	7mV	50nA	4V–32V		SOT-23-5	high-voltage general purpose
MIC6251	GainBlock™ Amplifier	2MHz	6V/μs	1.3mA	14mV	50nA	4V–32V		SOT-23-5	+2, +1, –1 selectable gain
MIC6252	GainBlock™ Amplifier	2MHz	6V/μs	1.3mA	7mV	50nA	4V–32V		SOT-23-5	+0.5, +1 selectable gain or average value
LMC7101	General Purpose Op Amp	500kHz	0.5V/μs	500μA	6mV	1pA	2.7V–10V	input/output	SOT-23-5	rail-to-rail input and output
MIC7111	Micropower 1.8V Op Amp	25kHz	20mV/μs	15μA	7mV	1pA	1.8V–11V	input/output	SOT-23-5	rail-to-rail input and output
MIC7122	Rail-to-Rail Dual Op Amp	750kHz	0.7V/μs	350μA	9mV	1pA	2.2V–15V	input/output	MSOP-8	rail-to-rail input and output
MIC7201	GainBlock™ Amplifier	400kHz	500mV/μs	600μA	44mV	n/a	2.2V–10V	input/output	SOT-23-5	rail-to-rail input and output
MIC7300	High Output Drive Op Amp	500kHz	500mV/μs	700μA	9mV	0.5pA	2.2V–10V	input/output	SOT-23-5 MSOP-8	rail-to-rail input and output
MIC910	135MHz Op Amp	135MHz	270mV/μs	2.4A	15mV	3.5μA	5V–18V		SOT23-5	low power/high speed
MIC911	105MHz Op Amp	105MHz	120mV/μs	1.25μA	10mV	1.5μA	5V–18V		SOT23-5	low power/high speed
MIC912	200MHz Op Amp	200MHz	360mV/μs	2.4μA	15mV	3.5μA	5V–18V		SOT23-5	low power/high speed
MIC913	350MHz Op Amp	350MHz	500mV/μs	4.2μA	16mV	5.5μA	5V–18V		SOT23-5	low power/high speed
MIC914	160MHz Op Amp	160MHz	160mV/μs	1.25μA	10mV	1.5μA	5V–18V		SOT23-5	low power/high speed
MIC915	Dual MIC910	135MHz	270mV/μs	2.4μA	15mV	3.5μA	5V–18V		MSOP-10	low power/high speed
MIC916	Triple MIC910	135MHz	270mV/μs	2.4μA	15mV	3.5μA	5V–18V		QSOP-8	low power/high speed

Comparator Selection Guide

Device	Description	Response Time	Supply Current	Input Offset Voltage (max.)	Input Bias Current	Supply Range	Package	Comments
MIC6270	High-Voltage Comparator	600ns	0.3mA	5mV	25nA	2V–36V	SOT-23-5	
MIC7211	Rail-to-Rail Input Comparator	4μs	5μA	10mV	5pA	2.2V–10V	SOT-23-5	push-pull output
MIC7221	Rail-to-Rail Input Comparator	4μs	5μA	10mV	5pA	2.2V–10V	SOT-23-5	open-drain output
MIC833	Comparator and Reference w/ Adjustable Hysteresis	5μs	1μA	n/a	5pA	1.5V–5.5V	SOT-23-5	internal 1% reference
MIC834	Comparator and Reference	5μs	1.5μA	n/a	5pA	1.5V–5.5V	SOT-23-5	internal 1% reference
MIC841	Comparator and Reference w/Adjustable Hysteresis	12μs	1.5μA	n/a	5pA	1.5V–5.5V	SC70-5	internal 1.25% reference
MIC842	Comparator and Reference	12μs	1.5μA	n/a	5pA	1.5V–5.5V	SC70-5	internal 1.25% reference

MOSFET Driver Selection Guide

Device	Function	Type	Logic	Peak Output	Output Impedance	On-Time (Delay+Rise Time)	Supply Voltage	Package	Comments
MIC426 see MIC4426	Low-Side Driver	Dual	Inverting	1.5A	6Ω	30ns into 1,000 pF	4.5V to 18V		Drives hex 0–hex 3 size MOSFET: 400pF to 3,000pF
MIC427 see MIC4427	Low-Side Driver	Dual	Noninverting	1.5A	6Ω	30ns into 1,000 pF	4.5V to 18V		Drives hex 0–hex 3 size MOSFET: 400pF to 3,000pF
MIC428 see MIC4428	Low-Side Driver	Dual	Inverting + Noninverting	1.5A	6Ω	30ns into 1,000 pF	4.5V to 18V		Drives hex 0–hex 3 size MOSFET: 400pF to 3,000pF
MIC1426 see MIC4426	Low-Side Driver	Dual	Inverting	1.2A	8Ω	38ns into 1,000pF	4.75V to 16V		Drives hex 0–hex 3 size MOSFET: 400pF to 3,000pF
MIC1427 see MIC4427	Low-Side Driver	Dual	Noninverting	1.2A	8Ω	38ns into 1000pF	4.75V to 16V		Drives hex 0–hex 3 size MOSFET: 400pF to 3,000pF
MIC1428 see MIC4428	Low-Side Driver	Dual	Inverting + Noninverting	1.2A	8Ω	38ns into 1,000 pF	4.75V to 16V		Drives hex 0–hex 3 size MOSFET: 400pF to 3,000pF
MIC4416	Low-Side Driver	Single	Noninverting	1.2A	3.5Ω	37ns into 1,000 pF	4.5V to 18V	SOT-143	IttyBitty® device
MIC4417	Low-Side Driver	Single	Inverting	1.2A	3.5Ω	37ns into 1,000 pF	4.5V to 18V	SOT-143	IttyBitty® device
MIC4420	Low-Side Driver	Single	Noninverting	6A	2.5Ω	20ns into 2,500pF	4.5V to 18V	SOP-8 MSOP-8 PDIP-8 TO-220-5 CerDIP-8(1)	Drives hex 6–hex 7 size MOSFET: 1,500pF to 16,000pF; Latch-up protected; input to –5V
MIC4429	Low-Side Driver	Single	Inverting	6A	2.5Ω	20ns into 2,500pF	4.5V to 18V	SOP-8 MSOP-8 PDIP-8 TO-220-5	Drives a hex 6–hex 7 size MOSFET: 1,500pF to 16,000pF; Latch-up protected; input to –5V
MIC4421	Low-Side Driver	Single	Inverting	9A	1Ω	25ns into 10,000pF	4.5V to 18V	PDIP-8 SOP-8 TO-220-5	Drives 1,500pF to 47,000pF; Latch-up protected; input to –5V
MIC4422	Low-Side Driver	Single	Noninverting	9A	1Ω	25ns into 10,000pF	4.5V to 18V	PDIP-8 SOP-8 TO-220-5	Drives 1,500pF to 47,000pF; Latch-up protected; input to –5V

Device	Function	Type	Logic	Peak Output	Output Impedance	On-Time (Delay+Rise Time)	Supply Voltage	Package	Comments
MIC4423	Low-Side Driver	Dual	Inverting	3A	3.5Ω	56ns into	4.5V to 18V 1,800pF	SOP-8 SOP-16 PDIP-8	Drives hex 4–hex 5 size MOSFET: 6,000pF to 12,000pF; Latch-up protected; input to –5V
MIC4424	Low-Side Driver	Dual	Noninverting	3A	3.5Ω	56ns into 1,800pF	4.5V to 18V	SOP-8 SOP-16 PDIP-8 CerDIP(2)	Drives a hex 4–hex 5 size MOSFET: 6,000pF to 12,000pF; Latch-up protected; input to –5V
MIC4425	Low-Side Driver	Dual	Inverting + Noninverting	3A	3.5Ω	56ns into 1,800pF	4.5V to 18V	SOP-8 SOP-16 PDIP-8	Drives a hex 4–hex 5 size MOSFET: 6,000pF to 12,000pF; Latch-up protected; input to –5V
MIC4426	Low-Side Driver	Dual	Inverting	1.5A	6Ω 1,000pF	35ns into	4.5V to 18V	SOP-8 MSOP-8 PDIP-8 CerDIP-8(3)	Drives a hex 0–hex 3 size MOSFET: 400pF to 3,000pF; Latch-up protected; input to –5V
MIC4427	Low-Side Driver	Dual	Noninverting	1.5A	6Ω 1,000pF	35ns into	4.5V to 18V	SOP-8 MSOP-8 PDIP-8 CerDIP-8(4)	Drives a hex 0–hex 3 size MOSFET: 400pF to 3,000pF; Latch-up protected; input to –5V
MIC4428	Low-Side Driver	Dual	Inverting + Noninverting	1.5A	6Ω	35ns into 1,000pF	4.5V to 18V	SOP-8 MSOP-8 PDIP-8 CerDIP-8(5)	Drives a hex 0–hex 3 size MOSFET: 400pF to 3,000pF; Latch-up protected; input to –5V
MIC4451	Low-Side Driver	Single	Inverting	12A	0.8Ω	35ns into	4.5V to 18V 15,000pF	SOP-8 PDIP-8 TO-220-5	Drives 1,500pF to 62,000pF; Latch-up protected; input to –5V
MIC4452	Low-Side Driver	Single	Noninverting	12A	0.8Ω	35ns into	4.5V to 18V 15,000pF	SOP-8 PDIP-8 TO-220-5	Drives 1,500pF to 62,000pF; Latch-up protected; input to –5V
MIC4467	Low-Side Driver	Quad	Noninverting NAND Inputs	1.2A	5Ω	44ns into 470pF	4.5V to 18V	SOP-16 wide PDIP-14	Drives a hex 0–hex 3 size MOSFET: 400pF to 3,000pF; Latch-up protected; input to –5V
MIC4468	Low-Side Driver	Quad	Noninverting AND Inputs	1.2A	5Ω	44ns into 470pF	4.5V to 18V	SOP-16 wide PDIP-14	Drives a hex 0–hex 3 size MOSFET: 400pF to 3,000pF; Latch-up protected; input to –5V
MIC4469	Low-Side Driver	Quad	Inverting + Noninverting AND Inputs	1.2A	5Ω	44ns into 470pF	4.5V to 18V	SOP-16 wide PDIP-14 CerDIP-14(6)	Drives a hex 0–hex 3 size MOSFET: 400pF to 3,000pF; Latch-up protected; input to –5V

Device	Function	Type	Logic	Peak Output	Output Impedance	On-Time (Delay+Rise Time)	Supply Voltage	Package	Comments
MIC5011	High- or Low-Side Driver	Single	Noninverting	–	–	60µs into 1,000pF	4.75V to 32V	SOP-8 PDIP-8	External charge-pump capacitors (opt.)
MIC5013	High- or Low-Side Driver	Single	Noninverting	–	–	60µs into 1,000pF	7V to 32V	SOP-8 PDIP-8	Current sense, fault
MIC5014	High- or Low-Side Driver	Single	Noninverting	–	–	90µs into 1,000pF	2.75V to 30V	SOP-8 PDIP-8	Low cost
MIC5015	High- or Low-Side Driver	Single	Inverting	–	–	90µs into 1,000pF	2.75V to 30V	SOP-8 PDIP-8	Low cost
MIC5016	High- or Low-Side Driver	Dual	Noninverting	–	–	90µs into 1,000pF	2.75V to 30V	SOP-16 wide PDIP-14	Low cost
MIC5017	High- or Low-Side Driver	Dual	Inverting	–	–	90µs into 1,000pF	2.75V to 30V	SOP-16 wide PDIP-14	Low cost
MIC5018	High- or Low-Side Driver	Single	Noninverting	–	–	2.1ms into 3,000pF	2.7V to 9V	SOT-143	IttyBitty® device
MIC5020	Complementary Low-Side Driver	Single	Noninverting	–	–	1.1µs into 1,500pF	11V to 50V	SOP-8 PDIP-8	Complement to MIC5021 (similar performance), current sense (50mV nominal)
MIC5021	High-Speed High-Side Driver	Single	Noninverting	–	–	0.9µs into 1,500pF	12V to 36V	SOP-8 PDIP-8	Current sense (50mV nominal)
MIC5022	Half-Bridge	Single	Noninverting High-Side	–	–	2.2µs into 1,500pF	12V to 36V	SOP-16 PDIP-14	Current sense (50mV nominal)

1. SMD (military) 5962-8877003PA.
2. SMD (military) 5962-8850305PA.
3. SMD (military) 5962-8850307PA.
4. SMD (military) 5962-8850308PA.
5. SMD (military) 5962-8850309PA.
6. SMD (military) 5962-9459403MCA.

Voltage Monitors and Voltage/Processor Supervisors Selection Guide

Standard Voltage Supervisors	Manual Reset Input	Reset Output Polarity	Reset Threshold Voltage	t _{RESET} (ms)	I _{SUPPLY} (μA)	Package
MIC1810-15/-10/-5	No	Low	4.12V, 4.37V, 4.62V	100	9.0	SOT23-3
MIC1815-20/-10	No	Low	2.55V, 2.88V	100	9.0	SOT23-3
MIC809R/S/T/J/M/L	No	Low	2.63V, 2.93V, 3.08V, 4.00V, 4.38V, 4.63V	140	9.0	SOT23-3
MIC810R/S/T/J/M/L	No	High	2.63V, 2.93V, 3.08V, 4.00V, 4.38V, 4.63V	140	9.0	SOT23-3
MIC811R/S/T/J/M/L	Yes	Low	2.63V, 2.93V, 3.08V, 4.00V, 4.38V, 4.63V	140	9.0	SOT-143
MIC812R/S/T/J/M/L	Yes	High	2.63V, 2.93V, 3.08V, 4.00V, 4.38V, 4.63V	140	9.0	SOT-143
MIC6315-xxD2	Yes	Low	2.5–5.0V in 100mV increments	20	9.0	SOT-143
MIC6315-xxD3	Yes	Low	2.5–5.0V in 100mV increments	140	9.0	SOT-143
MIC6315-xxD4	Yes	Low	2.5–5.0V in 100mV increments	1100	9.0	SOT-143

Standard Voltage Supervisors with Power Fail Detection	Manual Reset Input	Reset Output Polarity	Reset Threshold Voltage	t _{RESET} (ms)	I _{SUPPLY} (μA)	Package
MIC707	Yes	Both	4.65V	140	60	S0-8, PDIP-8
MIC708	Yes	Both	4.40V	140	60	S0-8, PDIP-8
MIC708R/S/T	Yes	Both	2.63V, 2.93V, 3.08V	140	30	S0-8, PDIP-8

Standard Voltage Supervisors for AMD Elan Microprocessors	Manual Reset Input	Reset Output Polarity	Reset Threshold Voltage	t _{RESET} (ms)	I _{SUPPLY} (μA)	Package
MIC8114T (SC400/410)	Yes	Low	3.08V	790	9.0	SOT-143
MIC8115T (SC500)	Yes	Low	3.08V	1100	9.0	SOT-143

Single and Dual Ultra-Low Voltage Supervisors

	Single/ Dual	Manual Reset Input	Reset Output(s) Polarity	Reset Threshold Voltage	t _{RESET} (ms)	I _{SUPPLY} (μ A)	Package
MIC2774H-xxBM5	Dual	Yes	High	Fixed: 1.69V, 2.25V, 2.34V, 2.53V	140	3.5	SOT23-5
MIC2774N-xxBM5			Open-Drain	2.67V, 2.81V, 2.93V, 3.09V, 4.43V, 4.68V;			
MIC2774L-xxBM5			Low	Adjustable from 0.300V min.			
MIC2775-xxBM5	Single	Yes	Both	1.69V, 2.25V, 2.34V, 2.53V, 2.67V, 2.81V 2.93V, 3.09V, 4.43V, 4.68V	140	3.5	SOT23-5
MIC2776N-BM5	Single	Yes	Open-Drain	Adjustable from 0.300V min.	140	3.5	SOT23-5
MIC2776L-BM5	Single	Yes	Low	Adjustable from 0.300V min.	140	3.5	SOT23-5
MIC2776H-BM5	Single	Yes	High	Adjustable from 0.300V min.	140	3.5	SOT23-5
MIC2777-xxBM5	Dual	Yes ⁽¹⁾	Both	Fixed: 1.69V, 2.25V, 2.34V, 2.53V 2.67V, 2.81V, 2.93V, 3.09V, 4.43V, 4.68V; Adjustable from 0.300V min.	140	3.5	SOT23-5

Voltage Monitors & Supervisors Optimized for Battery Monitoring

	Manual Reset Input	Reset Output(s) Polarity	V _{RESET}	Adjustable Hysteresis	t _{RESET} (ms)	I _{SUPPLY} (μ A)	Package
MIC833	No	High	Adjustable $\geq$ 1.25V	Y	n/a	1.0	SOT23-5
MIC834	No	High	Adjustable $\geq$ 1.25V	N	n/a	1.5	SOT23-5
MIC2755	Yes	Open-Drain	Adjustable $\geq$ 1.25V	Y	700	2.0	MSOP-8
MIC2778-1/-2 ⁽²⁾	No	Low	Adjustable $\geq$ 1.25V	Y	140	1.0	SOT23-5
MIC2779L-1/-2 ⁽²⁾	No	Low	Adjustable $\geq$ 1.25V	Y	140	1.0	SOT23-5
MIC2779H-1/-2 ⁽²⁾	No	High	Adjustable $\geq$ 1.25V	Y	140	1.0	SOT23-5
MIC841	No	High or Low	Adjustable $\geq$ 1.25V		n/a		SC70-5
MIC842	No	High or Low	Adjustable $\geq$ 1.25V		n/a		SC70-5

Microprocessor Supervisors with Watchdog Timers		Manual Reset Input	Reset Output(s) Polarity	Reset Threshold Voltage	t _{RESET} (ms)	Watchdog Timeout	Power Fail Detect	I _{SUPPLY} (μ A)	Package
MIC705		Yes	Low	4.65V	140	1.6s	Yes	60	SO-8, PDIP-8
MIC706		Yes	Low	4.40V	140	1.6s	Yes	60	SO-8, PDIP-8
MIC706P		Yes	High	2.63V	140	1.6s	Yes	30	SO-8, PDIP-8
MIC706R/S/T		Yes	Low	2.63V, 2.93V, 3.08V	140	1.6s	Yes	30	SO-8, PDIP-8
MIC1232		Yes	Both	4.62V or 4.37V	250	150ms/600ms/1.2s	Yes	30	SO-8, PDIP-8
MIC1832		Yes	Both	2.88V or 2.55V	250	150ms/600ms/1.2s	Yes	30	SO-8, PDIP-8

1. Manual reset switch connects to adjustable input.

2. -1 is 1% and -2 is 2% 1.25V reference tolerance.

Power Supply Supervisors w/Over-Voltage Monitoring		Reset Output	Crowbar Output	Typical Application Voltage	Package
MIC2753		Active High Output	Y	3.3V $\pm$ 5%, 3.3V $\pm$ 10%, 5V $\pm$ 5% 5V $\pm$ 10%, 2.85V $\pm$ 5%	SOT-23-5
MIC2754		Active Low Output	Y	3.3V $\pm$ 5%, 3.3V $\pm$ 10%, 5V $\pm$ 5% 5V $\pm$ 10%, 2.85V $\pm$ 5%	SOT-23-5

Latched Driver Selection Guide

Device	Input	Number Output Channels	Nominal Sink Current	Nominal Source Current	Maximum Output Voltage*	Thermal, UVLO, Overcurrent Protection	SOIC -40°C to +85°C	PLCC -40°C to +85°C	PDIP -40°C to +85°C	CerDIP -55°C to +125°C
MIC4807	Parallel 3-bit Address	8-bit	100mA	—	80V	■	—	—	18-pin	—
MIC5800	Parallel	4-bit	400mA	—	50V	—	14-pin	—	14-pin	—
MIC5801	Parallel	8-bit	400mA	—	50V	—	24-pin	28-pin	22-pin	22-pin†
MIC58P01	Parallel	8-bit	400mA	—	80V	■	24-pin wide	28-pin	22-pin	—
MIC5821	Serial	8-bit	400mA	—	50V / 35V	—	—	—	16-pin	—
MIC5822	Serial	8-bit	400mA	—	80V / 50V	—	—	—	16-pin	—
MIC5841	Serial	8-bit	400mA	—	50V / 35V	—	18-pin wide	20-pin	18-pin	—
MIC5842	Serial	8-bit	400mA	—	80V / 50V	—	18-pin wide	20-pin	18-pin	—
MIC58P42	Serial	8-bit	400mA	—	80V / 50V	■	18-pin wide	20-pin	18-pin	—
MIC5891	Serial	8-bit	—	400mA	50V	—	16-pin wide	—	16-pin	—
MIC59P50	Parallel	8-bit	400mA	—	80V	■	24-pin wide	28-pin	24-pin	—
MIC59P60	Serial	8-bit	400mA	—	80V / 50V	■	20-pin wide	20-pin	20-pin	—

* Second voltage, when shown, is for inductive load applications.

† SMD (military) 5962-8764001WA.

Thermal/System Management, Transistors and Arrays, Timers, References, and SCSI Terminators Selection Guide

2-Wire Serial Thermal Supervisors						
	Type	Remote Zone	Internal Zone	Data Width	Crit. Output	Devices Bus
MIC184	Local/Remote Thermal Supervisor	1***	Y	9	—	4(8)* SOP-8, MSOP-8
MIC284	2-Zone Thermal Supervisor w/CRIT output	1	Y	8	Y	2(8)** SOP-8, MSOP-8
MIC384	3-Zone Thermal Supervisor	2	Y	8	—	2(8)** SOP-8, MSOP-8
Fan Management ICs			Type	Package		
MIC502	Fan Management IC			DIP-8, SOP-8		
MIC74	2-Wire Serial I/O Expander and Fan Controller			QSOP-16		
I/O Expanders			Type	Package		
MIC74	2-Wire Serial I/O Expander and Fan Controller			QSOP-16		
Transistors		Type	Voltage Range	On-Resistance	Substrate Pin	Gate Pull-up Resistor
		Type				Package
MIC94030	Single P-Channel MOSFET	2.7V to 13.5V	0.75Ω @ V _{GS} = 4.5V	■		[4-pin] SOT-143
MIC94031	Single P-Channel MOSFET	2.7V to 13.5V	0.75Ω @ V _{GS} = 4.5V	■	■	[4-pin] SOT-143
Array		Type	Voltage	Input Signal		Package
MIC2981/82	Octal Source-Driver Array	50V	5V-TTL or 5V-to-15V CMOS or PMOS		18-pin DIP, 18-pin wide SOP	

Timers		Type	Package
MIC1555		IttyBitty® RC Timer/Oscillator	SOT-23-5
MIC1557		IttyBitty® RC Oscillator	SOT-23-5
References		Type	Package
LM4040/4041		Precision Micropower Shunt Voltage Reference	SOT-23
SCSI Terminators		Type	Package
MIC5204		SCSI-II Active Terminator	SOP-8, SOT-223

*In LM75 mode; 4 when using T1 function

**Alternate base addresses are available. Contact factory.

***MIC184 has limit registers for monitoring one zone at a time; either the internal or external zone may be monitored at any given time.

Display Driver Selection Guide

Device	Description	Segments	Package
MM5450	LED Display Driver	34	40-pin P-DIP 44-pin PLCC
MM5451	LED Display Driver	35	40-pin P-DIP 44-pin PLCC
MIC8030	50V LCD Driver	32	44-pin PLCC

QwikRadio™ Selection Guide

Device	Frequency Range	Sweep-Mode Data Rate	Fixed-Frequency Data Rate	Quiescent Current	Power Cycle	Package	Key Feature
MICRF001	300MHz–440MHz	4.8kb/s	4.8kb/s	6.3mA		SOP-14, DIP-14	lowest cost
MICRF011	300MHz–440MHz	2.5kb/s	10kb/s	2.4mA		SOP-14, DIP-14	lower power
MICRF002	300MHz–440MHz	2.5kb/s	10kb/s	2.4mA	■	SOP-16, DIP-16	power-cycle mode
MICRF022	300MHz–440MHz	2.5kb/s	10kb/s	2.4mA	■	SOP-8	smaller package
MICRF003	800MHz–1GHz	5kb/s	20kb/s	4mA	■	SOP-16	high frequency
MICRF033	800MHz–1GHz	5kb/s	20kb/s	4mA	■	SOP-8	smaller package
MICRF004	140MHz–200MHz	2.5kb/s	10kb/s	2.4mA	■	SOP-16, DIP-16	lower frequency
MICRF007	300MHz–440MHz	n/a	1.2kb/s	1.7mA	■	SOP-8	lowest power

Foundry Process Selection Guide

Overview

Micrel has offered foundry services since 1981 to IC designers and manufacturers seeking production solutions compatible with their specific application or technology needs. Micrel provides a variety of 4" and 6" wafer processing resources that address unique customer requirements.

Contact the Micrel Silicon Foundry Sales Manager at (408) 944-0800 for more information on foundry capabilities.

Service Options

- **Foundry:** Micrel duplicates client process
- **R&D Foundry:** Micrel develops new process to meet unique requirements
- **Semicustom:** Micrel technology, client design, Micrel process
- **Custom:** client circuit, Micrel technology, design, and process
- **Full Service:** client specification, Micrel design, technology, process, test, and packaging.

Process Summary

Process	Voltage	Feature Size	Description
Bipolar	1.5V to 150V	1.5μm to 8μm	junction-isolated bipolar $X_j = 3\mu\text{m}$, $3\Omega\text{-cm}$ to $5\Omega\text{-cm}$ epitaxial
CMOS	0.8V to 60V	1.5μm to 5μm	silicon-gate, self-aligned CMOS N-well or dual-well with optional isolation
BiCMOS	3.0V to 90V	1.5μm to 8μm	bipolar + 10μm metal gate CMOS
BCD2	3.0V to 40V	1.5μm to 3μm	CMOS + junction-isolated bipolar + DMOS
BCD3	3.0V to 60V	1.5μm to 3μm	CMOS + junction-isolated bipolar + DMOS
BCD5	3.0V to 120V	1.5μm to 8μm	CMOS + junction-isolated bipolar + DMOS

Communications Family

Micrel's communication products focus on the physical layer (PHY) of the high-speed networking and telecommunication markets. We supply products for high-speed serial data transmission and reception on copper and fiber optic mediums. Our products include PMD products for optical modules, transceivers for copper and optical connectivity, clock and data recovery products, and serializer/deserializer (SerDes) or mux/demux devices.

Leveraging the PLL intellectual property (IP) developed in the ClockWorks family, Micrel has built communication products with extremely low jitter. Our transceivers ease the problem of meeting the tight requirements specified for sending information across the network.

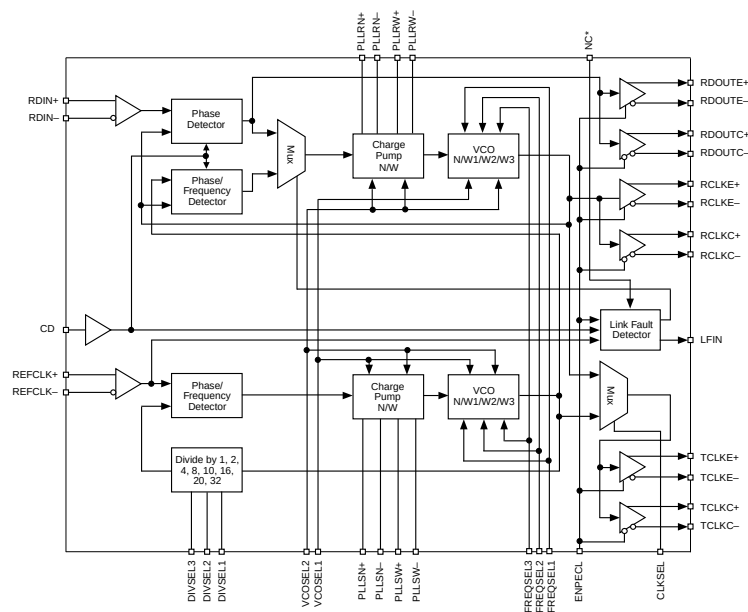
Micrel's silicon bipolar transceivers work at speeds up into the Gigabit-per-second range with clock and data recovery. They meet the stringent requirements imposed by the IEEE standards, giving our customers the absolute assurance that their systems are compliant. All of our SONET clock and data recovery and transceivers meet Bellcore jitter specifications of less than 0.01UI across the full wide-band spectrum—a capability unique in the industry.

SY87702L

3.3V 28Mbps–2.5Gbps AnyRate™ Clock and Data Recovery

Key Features

- A clock and data recovery unit with integrated clock multiplier unit capable of frequency agile continuous frequency functionality—covers both standard-specific and custom/proprietary rates
- 3.3V power supply
- Clock and data recovery and clock generation from 28Mbps up to 2.5Gbps (NRZ data stream for recovery)
- Differential PECL and CML outputs (selectable)
- Easy to use TTL select feature for CDR and CMU
- Low power
- Selectable reference frequencies
- Carrier detect input
- Lock detect output
- Line receiver input: no external buffering needed
- Industrial temperature
- Complies with Bellcore, ITU/CCITT, and ANSI specifications for SONET/SDH/ATM, IB, GbE, FC, 2X FC, ESCON, FICON, SDTV, HDTV, Firewire, and FDDI
- Available in 64-pin EPAD-TQFP

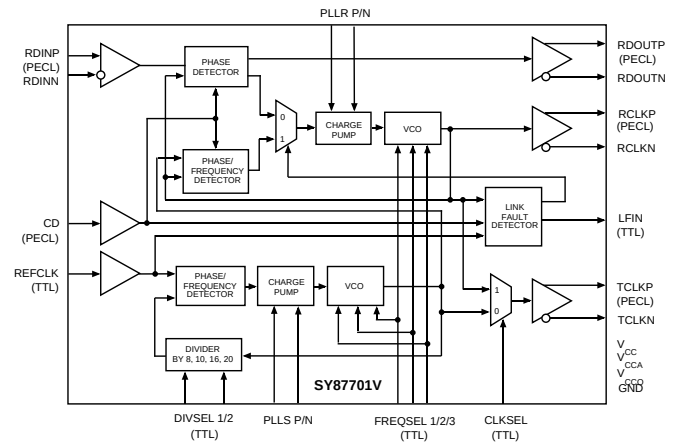


SY87701V

5V/3.3V 32Mbps–1.2Gbps AnyRate™ Clock and Data Recovery

Key Features

- A clock and data recovery unit with integrated clock multiplier capable of frequency agile functionality—covers both standard-specific and custom/proprietary rates
- 5V/3.3V power supply option
- Clock and data recovery and clock generation from 32Mbps up to 1.25Gbps (NRZ data stream for recovery)
- SONET/SDH/ATM, Fibre Channel, GbE, ESCON, and SMTE 259 compliant
- Two on-chip PLLs: (1) clock generation (2) clock recovery
- Selectable reference frequencies
- Line receiver input: no external buffering needed
- Complies with Bellcore, ITU/CCITT and ANSI specs for applications such as OC-1, OC-3, OC-12, ATM, FDDI, etc.
- Available in 32-pin EPAD-TQFP and 28-pin SOIC



SY87701L

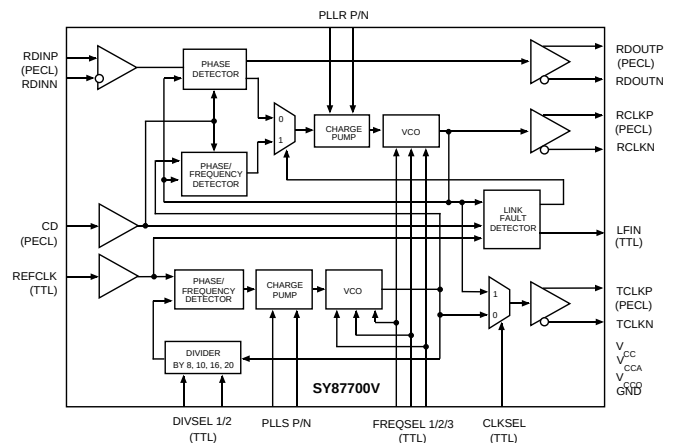
- Same as above
- 3.3V power supply option
- Industrial temperature range from -40°C to $+85^{\circ}\text{C}$

SY87700V

5V/3.3V 32-175Mbps AnyRate™ Clock and Data Recovery

Key Features

- A clock and data recovery unit with integrated clock multiplier capable of frequency agile functionality—covers both standard-specific and custom/proprietary rates
- 5V/3.3V power supply option
- Clock and data recovery and clock generation from 32Mbps up to 175Mbps (NRZ data stream for recovery)
- SONET/SDH/ATM compliant
- Two on-chip PLLs: (1) clock generation (2) clock recovery
- Selectable reference frequencies
- Line receiver input: no external buffering needed
- Complies with Bellcore, ITU/CCITT and ANSI specs for applications such as OC-1, OC-3, ATM, FDDI, etc.
- Available in 32-pin EPAD-TQFP and 28-pin SOIC



SY87700L

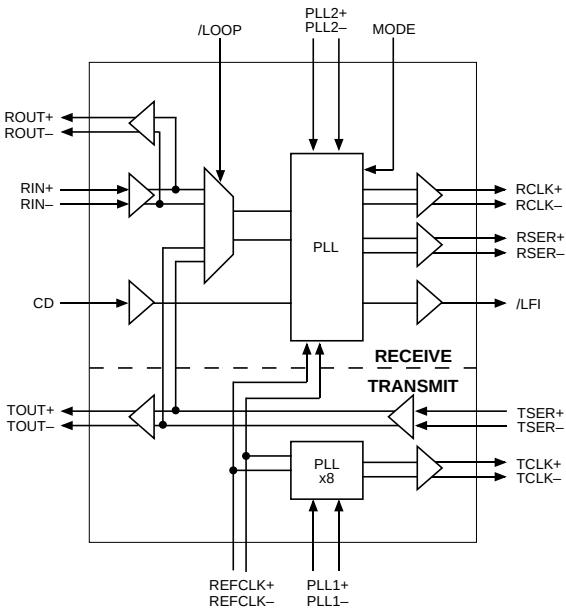
- Same as above
- 3.3V power supply option
- Industrial temperature range from -40°C to $+85^{\circ}\text{C}$

SY69952

OC-1/OC-3/STS-3 Clock and Data Recovery Transceivers

Key Features

- A complete single-chip transmitter and receiver
- Supports clock recovery and generation for 51.84Mbps or 155.52Mbps NRZ or NRZI data stream
- Complies with Bellcore, CCITT and ANSI specs
- Compatible with VLSI VN67200 and IgT WAC-413-A ATM Quad UNI Processors
- 28-pin SOIC

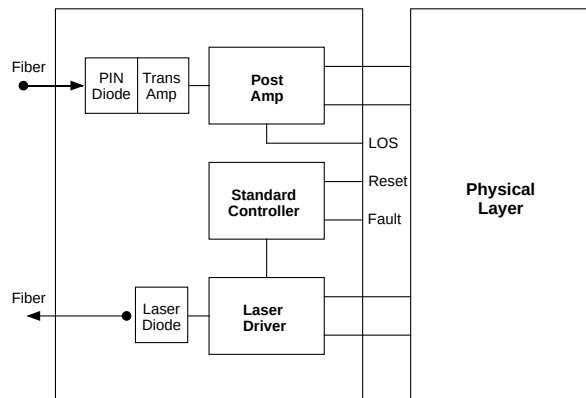


CDRs with Integrated Clock Synthesis

Part Number	Description	Output Range	Features
SY87700V	5V/3.3V 32-175Mbps AnyRate™ Clock and Data Recovery	32-175MHz	Two PLLs on-chip for clock generation and clock recovery
SY87700L	3.3V 32-175Mbps AnyRate™ Clock and Data Recovery	32-175MHz	Two PLLs on-chip for clock generation and clock recovery
SY87701V	5V/3.3V 32-1250Mbps AnyRate™ Clock and Data Recovery	32-1250MHz	Two PLLs on-chip for clock generation and clock recovery
SY87701L	3.3V 32-1250Mbps AnyRate™ Clock and Data Recovery	32-1250MHz	Two PLLs on-chip for clock generation and clock recovery
SY87702L	3.3V 28Mbps-2.5Gbps AnyRate™ Clock and Data Recovery	28Mbps-2.5Gbps	Two PLLs on-chip for clock generation and clock recovery

Fiber Optic Module

Micrel offers the best solution for Fiber Optic Transceiver modules because it provides the smallest footprint and the best performance at very low cost. The chipset provides the building blocks for laser driver, standard controller and post amplifier functions. All the chips are available in 3mm × 3mm MSOP. The optimized partition gives you low noise, a highly favorable “eye pattern” and a small footprint. This product family utilizes Micrel’s design expertise and ASSET™ process technology. The products work at speeds up to 3.125Gbps and V_{CC} down to 3.3V



Laser Drivers (Up to 3.125Gbps)

Part Number	Drive Type	Rate (Gbps)	V_{CC} (V)	Package
SY88902	AC/DC Coupled	Up to 1.25	5.0	10-pin MSOP
SY88922	AC/DC Coupled	Up to 2.5	5.0	10-pin MSOP
SY88922V	AC/DC Coupled	Up to 3.125	3.3/5.0	10-pin MSOP
SY88927V	AC Coupled	Up to 2.5	3.3/5.0	8-pin MSOP
SY100EL16V	AC Coupled	Up to 1.25	3.3/5.0	8-pin MSOP
SY100EL16VS	AC Coupled	Up to 1.25	3.3/5.0	8-pin MSOP

Post Amplifier — SY889X3 Family

Part Number	Rate (Gbps)	Output	V_{CC} (V)	LOS Output
SY88903V	Up to 1.5	PECL	3.3/5.0	TTL
SY88913V	Up to 1.5	PECL	3.3/5.0	PECL
SY88923V	Up to 2.5	PECL	3.3/5.0	TTL
SY88923AV	Up to 3.125	PECL	3.3/5.0	TTL
SY88943V	Up to 2.5	PECL	3.3/5.0	TTL
SY88993V	Up to 3.125	CML	3.3/5.0	TTL
SY88993AV	Up to 3.125	CML	3.3/5.0	TTL
SY88927V	Up to 2.5	PECL	3.3/5.0	NONE
SY100EL16V	Up to 1.25	PECL	3.3/5.0	NONE

Automatic Power Control (APC)

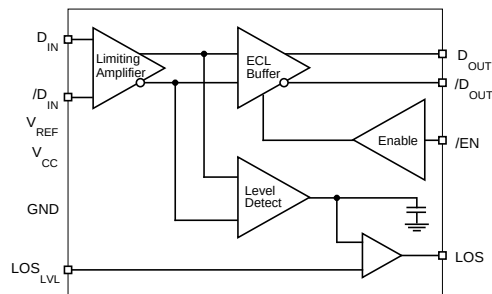
Part Number	Description
SY88905	Laser Diode Controller with APC

SY88903V

5V/3.3V 1.25Gbps High-Speed Limiting Post Amplifier

Key Features

- Up to 1.25Gbps operation
- Low noise
- Chatter-free /LOS generation
- Open-collector TTL LOS output
- TTL /EN input
- Differential PECL inputs for data
- Single power supply
- Designed for use with SY88905 and SY88922
- Tiny 10-pin MSOP (3mm)

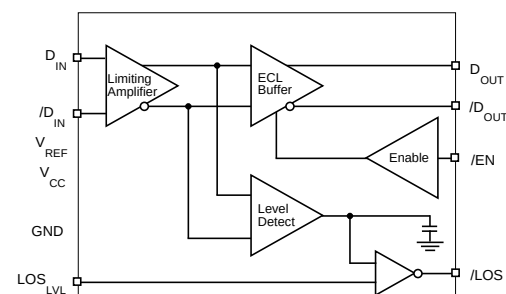


SY88913V

5V/3.3V 1.25Gbps High-Speed Limiting Post Amplifier

Key Features

- Up to 1.25Gbps operation
- Low noise
- Chatter-free /LOS generation
- PECL /LOS output
- TTL /EN input
- Differential PECL inputs for data
- /EN input is TTL with internal 27K pull up
- Single power supply
- Designed for use with SY88902 and SY88905
- Tiny 10-pin MSOP (3mm)

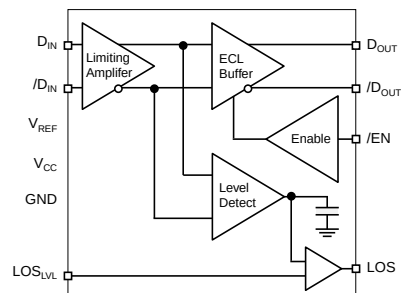


SY88923V/SY88923AV

5V/3.3V 3.2Gbps High-Speed Limiting Post Amplifier

Key Features

- Up to 3.2Gbps operation
- Same pinout as SY88903V
- Low noise
- Chatter-free LOS generation
- Open Collector TTL LOS Output
- TTL /EN input
- Differential PECL inputs for data
- 3.3V or 5V power supply
- Designed for use with Synergy™ laser diode driver and controller
- Tiny 10-pin MSOP (3mm)

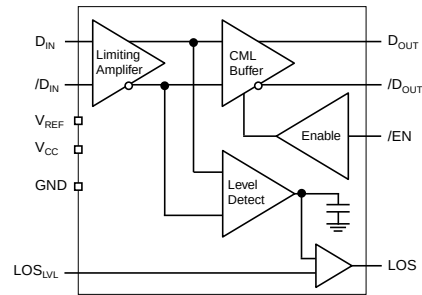


SY88993V/SY88993AV

5V/3.3V 2.5Gbps High-Speed Limiting Post Amplifier

Key Features

- Up to 3.2Gbps operation
- Low noise CML data outputs
- Chatter-free LOS generation
- Open collector TTL LOS output
- TTL /EN input
- Differential PECL inputs for data
- Single 3.3V or 5V power supply
- Chatter-free LOS generation
- Tiny 10-pin MSOP (3mm) and SOIC packages

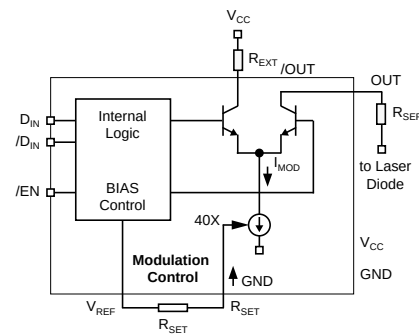


SY88922V

5V/3.3V 3.125Gbps Laser Driver

Key Features

- Up to 2.5Gbps operation
- 35mA modulation current
- Separate modulation control
- Separate output enable for laser safety
- Differential inputs for data
- 75K Ω input pulldown resistor
- Tiny 10-pin MSOP (3mm)

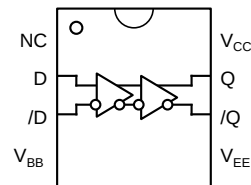


SY88927V

5V/3.3V 2.5Gbps High-Speed Differential Receiver

Key Features

- Maximum toggle frequency >3.0GHz
- Internal 75K Ω input pulldown resistor
- Faster version of SY100EL16V
- 100K ECL compatible I/O
- Improved output waveform characteristics
- Tiny 8-pin MSOP (3mm) and SOIC package

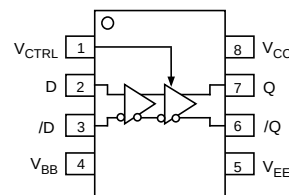


SY100EL16VS

5V/3.3V Variable Output Swing Differential Receiver

Key Features

- 3.3V and 5V power supply options
- High bandwidth output transitions
- Internal 75k Ω pulldown resistors on inputs
- Functionally equivalent to SY100EL16V with variable output swing
- Improved output waveform characteristics
- Tiny 8-pin MSOP (3mm) and SOIC packages



ClockWorks™ Family

ClockWorks™ products are easy-to-use standard products which make designing your high-frequency clock system a manageable task. They can be used to meet even the most stringent clock system requirements. ClockWorks are designed in Micrel's ASSET technology to provide plenty of drive for heavy loads and superior stability with minimal jitter and skew.

- **ClockWorks Frequency Synthesizers** generate the low-jitter, high-frequency clock signals required for high-speed systems, offering lower cost, higher quality and reliability than systems using expensive crystal oscillators.

- **ClockWorks Clock Distribution and Drivers** distribute high-frequency clock signals with very low skew in ECL, PECL or CMOS/TTL levels within 3.3V and/or 5V systems.

- **ClockWorks Clock Generators** divide or regenerate clock signals and lower frequency signals with minimal skew.

- For more critical timing problems and race conditions, **ClockWorks PLLs (Phase-Locked Loops)** can be used with clock distribution and driver devices to produce copies of phase coherent clock signals (i.e., zero phase-shift relative to inputs and outputs) and 2× frequency-multiplied clock signals.

Frequency Synthesizers

3.3V and/or 5V power supply
PECL/TTL/ECL
Low jitter
Superior noise immunity with differential VCO and external filter
Low power consumption
Ideal clock source for PECL/TTL/ECL systems
PLCC/MLCC, SOIC and EPAD-TQFP packages
Industrial temperature range*
(–40°C to +85°C)

Part Number	Description	Output Range	Features
SY89426	SONET OC-12/OC-3 Clock Synthesizer	622.08/155.52MHz	Retimed reference output
SY89429A	Programmable Frequency Synthesizer	25–400MHz	5V w/3.3V output option
SY89429V	5V/3.3V Programmable Frequency Synthesizer	25–400MHz	Programmable in 1MHz steps
SY89430V	5V/3.3V Programmable Frequency Synthesizer	50–950MHz	Programmable in 1MHz steps
SY87702L*	3.3V 28Mbps-2.5Gbps AnyRate™ Clock and Data Recovery	28Mbps–2.5Gbps	Two PLLs on-chip for clock generation and clock recovery
SY87701V	5V/3.3V 32-1250Mbps AnyRate™ Clock and Data Recovery	32–1250MHz	Two PLLs on-chip for clock generation and clock recovery
SY87701L*	3.3V 32-1250Mbps AnyRate™ Clock and Data Recovery	32–1250MHz	Two PLLs on-chip for clock generation and clock recovery
SY87700V	5V/3.3V 32-175Mbps AnyRate™ Clock and Data Recovery	32–175MHz	Two PLLs on-chip for clock generation and clock recovery
SY87700L*	3.3V 32-175Mbps AnyRate™ Clock and Data Recovery	32–175MHz	Two PLLs on-chip for clock generation and clock recovery

Phase-Locked Loops

3.3V and/or 5V power supply
PECL/TTL/ECL
Low-power differential design
2X frequency multiplication
External 2.5GHz VCO input capability
Ideal clock source for PECL/TTL/ECL systems
PLCC and SOIC packages

Part Number	Description	Input Range	Features
SY89420V	Dual Phase-Locked Loop w/Frequency Doubler Mode	30–560MHz	Optional 2X frequency multiplication
SY89421V	High-Performance Phase-Locked Loop	12–560MHz	External 2.5GHz VCO capability
SY89423V	Dual High-Performance Phase-Locked Loop	12–560MHz	External 2.5GHz VCO capability

Clock Generators

3.3V and/or 5V power supply
PECL/TTL/ECL

Low skew

Gigahertz toggle frequencies

SOIC package

Part Number	Description	Toggle Frequency	Features
SY10/100EL32V	+2 Divider	3.0GHz	510ps propagation delay
SY10/100EL33/L	+4 Divider	4.0GHz	650ps propagation delay
SY10/100EL34/L	+2, +4, +8 Clock Generation	1.1GHz	50ps output-to-output skew
SY10/100EL38/L	+2, +4/6 Clock Generation	1.0GHz	50ps output-to-output skew
SY100S834/L	(+1, +2, +4) or (+2, +4, +8) Clock Generation	1.1GHz	50ps output-to-output skew
SY100S838/L	(+1, +2/3) or (+2, +4/6) Clock Generation	1.0GHz	50ps output-to-output skew
SY100S839V	(+2/4, +4/5/6) Clock Generation	1.0GHz	50ps output-to-output skew

Clock Distribution and Drivers

3.3V and/or 5V power supply
PECL/TTL/ECL

Ultra-low skew; ideal for systems
requiring low skew clock networks

Differential internal designs for
superior noise immunity and stable
threshold inputs

Extra power/ground pins to reduce
crosstalk/noise

PLCC and SOIC packages

Part Number	Description	Outputs	Max Pin-to-Pin Skew (ps)
SY10/100E111	1:9 Differential Clock Driver w/Enable	9	50
SY10/100E111A/L	5V/3.3V 1:9 Differential Clock Driver without Enable	9	50
SY10/100E111AE/LE	5V/3.3V 1:9 Differential Clock Driver w/Enable	9	50
SY10/100EL11V	5V/3.3V 1:2 Differential Fanout Buffer	2	20
SY100EL14V	5V/3.3V 1:5 Clock Distribution	5	50
SY10/100EL15	1:4 Clock Distribution	4	50
SY100EL15L	3.3V 1:4 Clock Distribution	4	50
SY10/100ELT20V	5V/3.3V TTL-to-Differential PECL Translator	1	n/a
SY10/100ELT21	Differential PECL-to-TTL Translator	1	n/a
SY10/100ELT21L	3.3V Differential LVPECL-to-LVTTL Translator	1	n/a
SY10/100ELT23	Dual Differential PECL-to-TTL Translator	2	300
SY10/100ELT23L	3.3V Dual Differential PECL-to-TTL Translator	2	300
SY100EL29V	5V/3.3V Dual Differential Data and Clock D Flip-Flop w/Set and Reset	2	50
SY10/100H641	Single-Supply PECL-to-TTL 1:9 Clock Driver (135MHz)	9	300
SY10/100H641A	Single-Supply PECL-to-TTL 1:9 Clock Driver (80MHz)	9	300
SY10/100H641L	3.3V Single-Supply PECL-to-TTL 1:9 Clock Driver	9	300
SY100HA643	Enhanced Dual-Supply ECL-to-TTL 1:8 Clock Driver	8	400
SY10/100H646L	3.3V Single-Supply PECL/TTL-to-TTL 1:8 Clock Driver	8	300
SY10/100H841	Single-Supply PECL-to-TTL 1:4 Clock Driver w/Latched Output Enable	4	300
SY10/100H841L	3.3V Single-Supply PECL-to-TTL 1:4 Clock Driver w/Latched Output Enable	4	300
SY10/100H842	Single-Supply PECL-to-TTL 1:4 Clock Driver w/Output Enable	4	300
SY10/100H842L	3.3V Single-Supply PECL-to-TTL 1:4 Clock Driver w/Output Enable	4	300
SY100S811	1:9 PECL Clock Driver with Selectable TTL or PECL Clock Input	9	50
SY100S815	1:4 PECL Clock Driver with Selectable TTL or PECL Clock Input	4	50
SY89809L	3.3V 1:9 Differential HSTL Clock Driver	9	50
SY89824L	3.3V 1:22 Differential HSTL Clock Driver	22	50

Programmable Delay Lines/Timing Edge Verniers

15/20ps step delay for precise edge
placement

Up to 2/40ns delay range with
cascade circuitry for increased delay
range

125MHz to >1GHz bandwidth

Less than ± 1 LSB timing accuracy

Part Number	Description	Frequency	Features
SY10/100E195	Programmable Delay Chip	>1GHz	600mV output swing
SY10/100E196	Programmable Delay Chip w/Analog Input	>1GHz	Linear input for tighter resolution
SY604	Trigger Programmable Timing Edge Vernier	125MHz	Pin-for-pin replacement for Brooktree Bt604
SY605	Write Programmable Timing Edge Vernier	125MHz	Pin-for-pin replacement for Brooktree Bt605

Logic Family

Each of our logic families offer ultrafast gate speeds and a wide variety of functions. Micrel's are the only products that can be mixed and matched across families to meet your functionality and cost requirements.

- **The *ECLinPS* family:** over 50 full-function products packaged in 28-pin PLCCs.
- **The *ECLinPS Lite* family:** over 30 single-function products in tiny MSOP and SOIC packages.
- **The *ECL Pro™* family:** next generation ECL products in tiny MSOP and SOIC packages.
- **The *SuperLite™* family:** new CML logic family in tiny MSOP and QSOP packages.

- The ***Super-300K* family:** more than 30 medium-function products.
- Add Micrel's ***Translator* family** to complete your overall solution.

And we also offer equivalent 3.3V products for leading-edge, high-performance, low-voltage systems.

Many products are available in the industrial temperature range of -40°C to $+85^{\circ}\text{C}$, and the entire ECLinPS Lite and ECL Pro product families are industrial temperature range guaranteed.

ECLinPS

Ultrafast gate speed for performance-oriented PECL/TTL/ECL systems
100% compatible with ON Semiconductor

Typical edge rates of 500ps

Industry standard 10KH and 100K ECL power supply and I/O levels

Packaged in 28-pin PLCC

Extended supply voltage options:

$V_{EE} = -4.2\text{V}$ to -5.5V

$V_{EE} = -3.0\text{V}$ to -3.8V

Some products specified in Industrial temperature range (-40°C to $+85^{\circ}\text{C}$)

Part Number	Description	Max. Delay (ps)	10KH I_{EE} Max. (mA)	100K I_{EE} Max. (mA)
SY10/100E016	8-Bit Synchronous Binary Up Counter	700MHz*	181	208
SY10/100E101	Quad 4-Input OR/NOR Gate	500	36	42
SY10/100E104	Quint 2-Input AND/NAND Gate	600	46	53
SY10/100E107	Quint 2-Input XOR/XNOR Gate	600	50	58
SY10/100E111	1:9 Differential Clock Driver with Enable	630	60	69
SY10/100E111A/L	5V/3.3V 1:9 Differential Clock Driver without Enable	630	66	78
SY10/100E111AE/LE	5V/3.3V 1:9 Differential Clock Driver with Enable	630	66	78
SY10/100E112	Quad Driver	600	56	65
SY10/100E116	Quint Differential Line Receiver	450	35	40
SY10/100E122	9-Bit Buffer	500	49	57
SY10/100E131	4-Bit D Flip-Flop	700	70	81
SY10/100E136	6-Bit Universal Up/Down Counter	550MHz*	150	170
SY10/100E137	8-Bit Ripple Counter	1800MHz*	145	167
SY10/100E141	8-Bit Shift Register	975	157	181
SY10/100E142	9-Bit Shift Register	1000	145	165
SY10/100E143	9-Bit Hold Register	1000	145	165
SY10/100E150	6-Bit D Latch	550	62	72
SY10/100E151	6-Bit D Register	800	78	90
SY10/100E154	5-Bit 2:1 Mux-Latch	700	91	105
SY10/100E155	6-Bit 2:1 Mux-Latch	700	102	117
SY10/100E156	3-Bit 4:1 Mux-Latch	900	90	103
SY10/100E157	Quad 2:1 Multiplexer	550	40	46
SY10/100E158	5-Bit 2:1 Multiplexer	550	40	46
SY10/100E160	12-Bit Parity Generator/Checker	950	98	113
SY10/100E163	2-Bit 8:1 Multiplexer	800	88	100
SY10/100E164	16:1 Multiplexer	850	71	81
SY10/100E166	9-Bit Magnitude Comparator	1100	136	156
SY10/100E167	6-Bit 2:1 Mux-Register	800	113	130
SY10/100E171	3-Bit 4:1 Multiplexer	650	67	77
SY10/100E175	9-Bit Latch with Parity	800	132	152

*Minimum guaranteed maximum operating frequency

Part Number	Description	Max. Delay (ps)	10KH I _{EE} Max. (mA)	100K I _{EE} Max. (mA)
SY10/100E193	Error Detection/Correction Circuit	1000	134	155
SY10/100E195	Programmable Delay Chip	1765	156	179
SY10/100E196	Programmable Delay Chip w/Analog Input	1756	156	179
SY10/100E212	3-Bit Scannable Register	1025	96	110
SY10/100E241	8-Bit Scannable Register	975	150	173
SY10/100E256	3-Bit 4:1 Mux-Latch	900	83	96
SY10/100E336	3-Bit Registered Bus Transceiver	1000	150	173
SY10/100E337	3-Bit Scannable Registered Bus Transceiver	1000	174	200
SY10/100E404	Quad Differential AND/NAND	650	127	146
SY10/100E416	Quint Differential Line Receiver	500	162	186
SY100E417	Quint LVPECL-to-PECL or PECL-to-LVPECL Translator	600	n/a	186
SY10/100E431	3-Bit Differential Flip-Flop	750	132	152
SY10/100E445	4-Bit Serial-to-Parallel Converter	2000MHz*	185	212
SY10/100E446	4-Bit Parallel-to-Serial Converter	1300MHz*	132	152
SY10/100E451	6-Bit Register Differential Data Clock	800	101	116
SY10/100E452	5-Bit Differential Register	800	89	102
SY10/100E457	Triple Differential 2:1 Multiplexer	650	110	127

*Minimum guaranteed maximum operating frequency

ECLinPS Lite

Ultrafast gate speed for performance-oriented PECL/TTL/ECL systems

100% compatible with ON Semiconductor

Sub-500ps performance

Industry standard 10KH and 100K ECL power supply and I/O levels

Packaged in space-efficient high-performance SOIC and MSOP

Extended supply voltage options:

$V_{EE} = -4.2V$ to $-5.5V$

$V_{EE} = -3.0V$ to $-3.8V$

Most products specified over industrial temperature range ($-40^{\circ}C$ to $+85^{\circ}C$)

Part Number	Description	Propagation delay (ps)	10KH I_{EE} Max. (mA)	100K I_{EE} Max. (mA)
SY10/100EL01	4-Input OR/NOR	350	17	20
SY10/100EL04	2-Input AND/NAND	395	17	20
SY10/100EL05	2-Input Differential AND/NAND	420	22	25
SY10/100EL07	2-Input XOR/XNOR	415	17	20
SY10/100EL11V	5V/3.3V 1:2 Differential Fanout Buffer	365	31	36
SY10/100EL12	Low-Impedance Driver	480	17	20
SY100EL14V	5V/3.3V 1:5 Clock Distribution	880	n/a	40
SY10/100EL15	1:4 Clock Distribution	720	35	40
SY100EL15L	3.3V 1:4 Clock Distribution	720	n/a	35
SY10/100EL16V	5V/3.3V Differential Receiver	355	22	26
SY10/100EL16VA	5V/3.3V Differential Receiver w/High Gain	730	40	46
SY10/100EL16VB	EL16VA w/Extra QB output	730	40	46
SY10/100EL16VC	EL16VB w/Enable input	730	40	46
SY10/100EL16VD	All options (VA-VF) packaged in SOIC or MSOP	730	40	46
SY10/100EL16VE	EL16VB w/Differential Q, QB Output (no V_{BB})	730	40	46
SY10/100EL16VF	EL16VC w/Differential Data Input	730	40	46
SY100EL16VS	5V/3.3V Variable Output Swing Differential Receiver	250	n/a	26
SY100EL17V	5V/3.3V Quad Differential Receiver	580	n/a	31
SY100EL29V	5V/3.3V Dual Differential Data and Clock D Flip-Flop w/Set and Reset	740	n/a	50
SY10/100EL31	D Flip-Flop with Set & Reset	2200MHz*	38	38
SY10/100EL32V	5V/3.3V +2 Divider	2600MHz*	30	35
SY10/100EL33/L	5V/3.3V +4 Divider	3800MHz*	33	37
SY10/100EL34/L	5V/3.3V +2, +4, +8 Clock Generation Chip	1200	49	54
SY10/100EL35	JK Flip-Flop	1800MHz*	32	37
SY10/100EL38/L	5V/3.3V +2, +4/6 Clock Generation Chip	1300	65	75
SY10/100EL51	Differential Clock D Flip-Flop	2200MHz*	29	36
SY10/100EL52	Differential Data and Clock D Flip-Flop	2200MHz*	25	29
SY100EL56V	5V/3.3V Dual Differential 2:1 Multiplexer	540	n/a	24
SY10/100EL57/L	4:1 Differential Multiplexer	640	20	24
SY10/100EL58	2:1 Multiplexer	360	17	19
SY10EL89	Coaxial Cable Driver	490	28	n/a
SY100EL90V	5V/3.3V Triple ECL-to-PECL Translator	500	24	n/a
SY100EL91	5V Triple PECL-to-ECL Translator	620	n/a	22
SY100EL91L	3.3V Triple LVPECL-to-LVPECL Translator	620	n/a	22
SY100EL92	Triple PECL-to-LVPECL Translator	500	n/a	12
SY100EL1001	Laser Diode Driver with Input D Flip-Flop	1000	n/a	25
SY100EL1003	Laser Diode Driver with Output Enable	1000	n/a	25
SY10EL1189	Fibre Channel Coaxial Cable Driver	425	47	n/a
SY10/100ELT20V	5V/3.3V TTL-to-Differential PECL Translator	300	20	20
SY10/100ELT21	Differential PECL-to-TTL Translator	3.0	20	20
SY10/100ELT21L	3.3V Differential LVPECL-to-LVTTL Translator	2.0	20	20
SY10/100ELT22	Dual TTL-to-Differential PECL Translator	600	25	25
SY10/100ELT22L	3.3V Dual TTL-to-Differential PECL Translator	600	25	25
SY10/100ELT23	Dual Differential PECL-to-TTL Translator	3000	25	25
SY10/100ELT23L	3.3V Dual Differential PECL-to-TTL Translator	3000	30	30
SY100ELT24	TTL-to-Differential ECL Translator	900	n/a	20
SY100ELT25	Differential ECL-to-TTL Translator	3600	n/a	14
SY100ELT982	1.25Gbps PECL-to-CML Translator with Output Enable	480	n/a	25

*Minimum guaranteed maximum operating frequency

ECL Pro™

Ultrafast gate speed for performance-oriented PECL/TTL/ECL systems

100% compatible with ON Semiconductor

Typical edge rates of 300ps

Packaged in space-efficient high-performance SOIC and MSOP

Extended supply voltage options:

$V_{EE} = -4.2V$ to $-5.5V$

$V_{EE} = -3.0V$ to $-3.8V$

All products specified over industrial temperature range ($-40^{\circ}C$ to $+85^{\circ}C$)

Part Number	Description	Propagation delay (ps)	10KH I_{EE} Max. (mA)	100K I_{EE} Max. (mA)
SY10EP01V	5V/3.3V 4-Input OR/NOR	230	31	n/a
SY10EP16V	5V/3.3V High-Speed Differential Receiver	180	42	n/a
SY10EP58V	5V/3.3V 2:1 Multiplexer	275	40	n/a
SY10/100EPT20V	5V/3.3V LVTTTL/LVCMOS-to-Differential LVPECL Translator	300	20	20
SY100EPT21L	3.3V Differential LVPECL-to-LVTTTL Translator	1900	n/a	20
SY100EPT22V	5V/3.3V Dual LVTTTL/LVCMOS-to-Differential LVPECL Translator	300	n/a	25
SY100EPT23L	3.3V Dual Differential LVPECL-to-LVTTTL Translator	1900	n/a	30

SuperLite™

Ultrafast gate speed for performance-oriented PECL/LVPECL/CML systems

Typical edge rates of 130ps

Wide voltage range of 2.3V to 5.7V

Packaged in space-efficient high-performance MSOP package

Low power < 100mA

All products specified over industrial temperature range ($-40^{\circ}C$ to $+85^{\circ}C$)

Part Number	Description	t_r/t_f Max. (ps)	I_{EE} Max. (mA)
SY55851U	AnyGate™ CML Logic Chip	110	40
SY55852U	D Flip-Flop	150	36

Super-300K™

Ultrafast gate speeds for performance-oriented PECL/TTL/ECL systems

Pin-compatible with National/Fairchild and Signetics F100K and 300 series

Functions at almost twice the speed and half the power of F100K from National and Signetics

Almost 2X the speed at similar power of 300 series

Typical edge rates of 500ps

Industry standard 100K ECL power supply and I/O levels

Packaged in CERPACK and PLCC

Extended supply voltage options:

$V_{EE} = -4.2V$ to $-5.5V$

$V_{EE} = -3.0V$ to $-3.8V$

Many products specified over industrial temperature range ($-40^{\circ}C$ to $+85^{\circ}C$)

Part Number	Description	Max. Speed (ps)		IEE Max. (mA)
		Flatpack	PLCC	
SY100S301	Triple 5-Input OR/NOR Gate	800	750	25
SY100S302	Quint 2-Input OR/NOR Gate	750	700	45
SY100S304	Quint AND/NAND Gate	1150	1050	60
SY100S307	Quint Exclusive OR/NOR Gate	1100	1000	58
SY100S313	Quad Driver	850	800	60
SY100S314	Quint Differential Line Receiver	1000	900	60
SY100S317	Triple 2-Wide OA/OAI Gate	1000	900	48
SY100S318	5-Wide 5, 4, 4, 2 OA/OAI Gate	900	800	55
SY100S321	Low-Power 9-Bit Inverter	800	700	55
SY100S322	9-Bit Buffer	800	700	55
SY100S324	Low-Power Hex TTL-to-ECL Translator	1400	1400	70
SY100S325	Low-Power Hex ECL-to-TTL Translator	2900	2900	37
SY100S331	Triple D Flip-Flop	800	700	80
SY100S336	4-Stage Counter/Shift Register	1200	1100	170
SY100S336A	Enhanced 4-Stage Counter/Shift Register	1200	1100	170
SY100S341	8-Bit Shift Register	1100	1000	150
SY100S350	Hex D Latch	1000	900	98
SY100S351	Hex D Flip-Flop	1200	1200	98
SY100S355	Quad Multiplexer/Latch	1200	1100	80
SY100S360	Dual Parity Checker/Generator	2300	2200	70
SY100S363	Dual 8-Input Multiplexer	1000	900	92
SY100S364	16-Input Multiplexer	1400	1300	63
SY100S366	9-Bit Comparator	1600	1500	120
SY100S370	Universal Demultiplexer/Decoder	1300	1200	92
SY100S371	Triple 4-Input Multiplexer w/Enable	1100	1000	68
SY100S391	Low-Power Hex TTL-to-PECL Translator	1400	1400	69
SY100S811	1:9 PECL Clock Driver with Selectable TTL or PECL Clock Input	—	950	74
SY100S815	1:4 PECL Clock Driver with Selectable TTL or PECL Clock Input	—	—	74
SY100S834/L	(+1, +2, +4) or (+2, +4, +8) Clock Generation Chip	—	—	54
SY100S838/L	(+1, +2/3) or (+2, +4/6) Clock Generation Chip	—	—	75
SY100S863	8-Input PECL Differential Mux w/TTL Selects	—	900	88
SY100S891	5-Bit Registered Transceiver	—	1500	216

Translators

High speed; low power

Choice of bus width and package type for flexibility

Single or dual supply translation

Suitable for PECL/TTL/ECL/CML/HSTL/LVDS systems

Part Number	Description	In/Output Translation		
		Bus Width	Speed (ns)	Single/Differential
SY10H350	PECL-to-TTL Translator	4	3.5	Differential In/Single Out
SY10H351	TTL/NMOS-to-PECL Translator	4	1.3	Single In/Differential Out
SY10H352	CMOS-to-PECL Translator	4	1.3	Single In/Differential Out
SY10/100H600	TTL-to-ECL w/TTL, ECL Enable	9	3.5	Single In/Single Out
SY10/100H601	ECL-to-TTL w/3-State Enable	9	4.8	Single In/Single Out
SY10/100H602	Latched TTL-to-ECL	9	3.5	Single In/Single Out
SY10/100H603	Latched ECL-to-TTL	9	4.8	Single In/Single Out
SY10/100H606	Registered Hex TTL-to-PECL	6	3.0	Single In/Differential Out
SY10/100H607	Registered Hex PECL-to-TTL	6	6.5	Differential In/Single Out
SY10/100H641	Single-Supply 1:9 PECL-to-TTL (135MHz)	1:9	6.3	Differential In/Single Out
SY10/100H641A	Single-Supply 1:9 PECL-to-TTL (80MHz)	1:9	6.3	Differential In/Single Out
SY10/100H641L	3.3V Single-Supply PECL-to-TTL	1:9	6.3	Differential In/Single Out
SY100HA643	Dual-Supply Octal ECL-to-TTL	1:8	3.1	Differential In/Single Out
SY10/100H646L	3.3V Single-Supply Octal PECL/TTL-to-TTL	1:8	6.5	Differential In/Single Out
SY10/100H841	Single-Supply Quad PECL-to-TTL w/Latched Output Enable	1:4	3.7	Differential In/Single Out
SY10/100H841L	3.3V Single-Supply Quad PECL-to-TTL w/Latched Output Enable	1:4	3.7	Differential In/Single Out
SY10/100H842	Single-Supply Quad PECL-to-TTL w/Output Enable	1:4	3.5	Differential In/Single Out
SY10/100H842L	3.3V Single-Supply Quad PECL-to-TTL w/Output Enable	1:4	3.5	Differential In/Single Out
SY100S324	Low-Power Hex TTL-to-ECL	6	1.5	Single In/Differential Out
SY100S325	Low-Power Hex ECL-to-TTL	6	3.0	Differential In/Single Out
SY100S391	Low-Power Hex TTL-to-PECL	6	1.4	Single In/Differential Out
SY100S811	Single-Supply PECL/TTL-to-PECL	9	0.63	Single or Differential In/Differential Out
SY100S815	Single-Supply Quad PECL/TTL-to-PECL	4	0.63	Single or Differential In/Differential Out
SY100E417	Quint LVPECL-to-PECL or PECL-to-LVPECL	5	0.5	Differential In/Differential Out
SY100EL90V	5V/3.3V Triple ECL-TO-PECL Translator	3	0.5	Differential In/Differential Out
SY100EL91	5V Triple PECL-TO-ECL Translator	3	0.62	Differential In/Differential Out
SY100EL91L	3.3V Triple LVPECL-TO-LVECL Translator	3	0.62	Differential In/Single Out
SY100EL92	Triple PECL-TO-LVPECL Translator	3	0.61	Differential In/Differential Out
SY10/100EPT20V	5V/3.3V TTL-to-Differential PECL Translator	1	0.3	Single In/Differential Out
SY100EPT21L	3.3V Differential LVPECL-to-LVTTL Translator	1	1.9	Differential In/Single Out
SY100EPT22V	5V/3.3V Dual LVTTL-to-Differential LVPECL Trans.	2	0.3	Single In/Differential Out
SY100EPT23L	3.3V Dual Differential LVPECL-to-LVTTL Trans.	2	1.9	Differential In/Single Out
SY10/100ELT20V	5V/3.3V TTL-to-Differential PECL Translator	1	0.3	Single In/Differential Out
SY10/100ELT21	Differential PECL-to-TTL Translator	1	2.5	Differential In/Single Out
SY10/100ELT21L	3.3V Differential LVPECL-to-LVTTL Translator	1	2	Differential In/Single Out
SY10/100ELT22	Dual TTL-to-Differential PECL Translator	2	1.1	Single In/Differential Out
SY10/100ELT22L	3.3V Dual LVTTL-to-Differential LVPECL Trans.	2	1.1	Single In/Differential Out
SY10/100ELT23	Dual Differential PECL-to-TTL Translator	2	3.5	Differential In/Single Out
SY10/100ELT23L	3.3V Dual Differential LVPECL-to-LVTTL Trans.	2	3.5	Differential In/Single Out
SY100ELT24	TTL-to-Differential ECL Translator	1	0.5	Single In/Differential Out
SY100ELT25	Differential ECL-to-TTL Translator	1	2.6	Differential In/Single Out
SY100ELT982	PECL-to-CML Translator with Output Enable	1	1.0	Differential In/Differential Out
SY55851U	2:1 PECL/LVPECL-to-CML	2:1	0.35	Differential In/Differential Out
SY89809L	3.3V PECL-to-HSTL Translator	1:9	1	Differential In/Differential Out
SY89824L	3.3V PECL-to-HSTL Translator	1:22	1	Differential In/Differential Out
SY88922	2.5Gbps PECL-to-CML Translator with Enable	1	1.0	Differential In/Differential Out

Analog Cross Reference

Allegro/Sprague

Part Number	Direct Cross	Functional Cross	Comments
A8184		MIC5207	
A8187		MIC5207	
A8188		MIC5209	
UCN4807		MIC4807	
UCN5800	MIC5800		
UCN5801	MIC5801		
UCN5810	MIC5810		
UCN5812	MIC5812		
UCN5818	MIC5818		
UCN5821	MIC5821		
UCN5822	MIC5822		
UCN5841	MIC5841		
UCN5842	MIC5842		
UCN5891	MIC5891		
UDN2981	MIC2981		
UDN2982	MIC2982		

Analog Devices

Part Number	Direct Cross	Functional Cross	Comments
ADM705	MIC705		
ADM706/P/R/S/T	MIC706/P/R/S/T		
ADM707	MIC707		
ADM708/R/S/T	MIC708/R/S/T		
ADM809	MIC809		
ADM810	MIC810		
ADM811	MIC811		
ADM812	MIC812		
ADM1232	MIC1232		
ADM1810	MIC1810		
ADM1815	MIC1815		
ADM6315	MIC6315		
ADP3300		MIC5205	
ADP3301		MIC2950	
ADP3302		MIC5202	
ADP3303		MIC5201	
ADP3306		MIC2920	
ADP3307		MIC5205	
ADP3308	MIC5246		
ADP3309	MIC5246		
ADP3330		MIC5245	
ADP3335		MIC5209	
ADP3338		MIC39100	
ADP3820		MIC79050	
AD8022		MIC911	
AD8041		MIC914	
AD8055/6		MIC913/MIC914	
AD8057/8		MIC913/MIC911	
AD8061/2/3		MIC913/MIC910	
AD817/847		MIC910	

Analog Microelectronics

Part Number	Direct Cross	Functional Cross	Comments
AME1084		MIC29502	
AME1085		MIC29302	
AME1117		MIC39100/1/2	
AME8800		MIC5245	
AME8801	MIC5245		
AME8802	MIC5246		
AME8805/6		MIC5209	
AME8807	MIC5209		
AME8810		MIC5209	
AME8815		MIC39150/1	
AME8816	MIC39151		
AME8820		MIC5245	

Calogic

Part Number	Direct Cross	Functional Cross	Comments
CLM1117		MIC39100	
CLM2805	MIC5205/5245		
CLM2810		MIC39102	
CLM2815		MIC2915X/MIC3915X	
CLM2830		MIC2930X	
CLM2850		MIC2950X	
CLM2930/1		MIC5236	
CLM2950	MIC2950/1		
CLM5205	MIC5205		
CLM4420	MIC4420		
CLM4429	MIC4429		
CLM4426	MIC4426		
CLM4427	MIC4427		
CLM4428	MIC4428		
LP2950	LP2950		
LP2951	LP2951		

Dallas Semiconductor (now parr of Maxim)

Part Number	Direct Cross	Functional Cross	Comments
DS1232	MIC1232		
DS1232LP	MIC1232		
DS1705	MIC705		
DS1706/P/R/S/T	MIC706/P/R/S/T		
DS1707	MIC707		
DS1708/R/S/T	MIC708/R/S/T		
DS1810	MIC1810		
DS1815	MIC1815		
DS1832	MIC1832		

élantec

Part Number	Direct Cross	Functional Cross	Comments
EL2044		MIC910/MIC912	
EL2244/2444		MIC910/MIC912	
EL5210		MIC913	
EL7104		MIC4420	
EL7114		MIC4429	
EL7202		MIC4424	

élantec (continued)

Part Number	Direct Cross	Functional Cross	Comments
EL7212		MIC4423	
EL7222		MIC4425	

EM Microelectronics – Marin S.A.

Part Number	Direct Cross	Functional Cross	Comments
V6309		MIC809	
V6319		MIC810	

IMP

Part Number	Direct Cross	Functional Cross	Comments
IMP705	MIC705		
IMP706/P/R/S/T	MIC706/P/R/S/T		
IMP707	MIC707		
IMP708/R/S/T	MIC708/R/S/T		
IMP809	MIC809		Minor spec. differences, see data sheets
IMP810	MIC810		Minor spec. differences, see data sheets
IMP811	MIC811		Minor spec. differences, see data sheets
IMP812	MIC812		Minor spec. differences, see data sheets
IMP1232LP	MIC1232		
IMP1810	MIC1810		
IMP1815	MIC1815		
IMP1832	MIC1832		

Impala Linear

Part Number	Direct Cross	Functional Cross	Comments
ILC705	MIC705		
ILC706	MIC706		
ILC707	MIC707		
ILC708	MIC70		
ILC809	MIC809		
ILC810	MIC810		
ILC811	MIC811		
ILC812	MIC812		
ILC1232	MIC1232		
ILC1832	MIC1832		

Infineon (Siemens)

Part Number	Direct Cross	Functional Cross	Comments
TDA5200		MICRF002/MICRF003	

Intersil (formerly Harris Semiconductor)

Part Number	Direct Cross	Functional Cross	Comments
ICL7555		MIC1555	
ICL7660		MIC2660	
ICL7667	MIC4426		

Linear Technology

Part Number	Direct Cross	Functional Cross	Comments
LM185		LM4040/LM4041	
LM285		LM4040/LM4041	
LM385		LM4040/LM4041	

Linear Technology (continued)

Part Number	Direct Cross	Functional Cross	Comments
LT1071		MIC2171	
LT1072		MIC2172	
LT1083		MIC29750	
LT1084		MIC29500	
LT1085		MIC29300	
LT1086		MIC29150	
LT1117		MIC39100/1/2	
LT1120		MIC5206	
LT1121	MIC5201		
LT1129	MIC39100	MIC39100/1/2	
LT1160		MIC5022	
LT1171		MIC2171	
LT1172		MIC2172	
LT1185		MIC29302	
LT1188		MIC2505	
LT1232		MIC1232	
LT1241		MIC38C45/MIC38HC45	Improved alternative
LT1242		MIC38C42/MIC38HC42	Improved alternative
LT1243		MIC38C43/MIC38HC43	Improved alternative
LT1244		MIC38C44/MIC38HC44	Improved alternative
LT1245		MIC38C45/MIC38HC45	Improved alternative
LT1301		MIC3172	
LT1303		MIC2570	
LT1307		MIC2571	
LT1435		MIC2182	
LT1521		MIC5236	
LT1528		MIC29312	
LT1529		MIC29300	
LT1575		MIC5156	
LT1580		MIC29712	
LT1584		MIC29710	
LT1585		MIC29510	
LT1587		MIC310	
LT1761	MIC5245		
LT1762	MIC5245		
LT1763	MIC5209		
LT1764	MIC39300		
LT1806		MIC913/MIC911	
LT1809		MIC912	
LT1812		MIC913/MIC911	
LT1810		MIC913/MIC911	
LT1962		MIC5219	
LT1963		MIC3910X/3915X	
LTC1155		MIC5016	

Linfinity (Micro Semi)

Part Number	Direct Cross	Functional Cross	Comments
LX1552	MIC38C42/MIC38HC42		Improved alternative
LX1553	MIC38C43/MIC38HC43		Improved alternative
LX1554	MIC38C44/MIC38HC44		Improved alternative
LX1555	MIC38C45/MIC38HC45		Improved alternative
LX8020		MIC5201	

Linfinity (continued)

Part Number	Direct Cross	Functional Cross	Comments
LX8117/A		MIC39100/1/2	
LX8117/B		MIC3915X	
LX8383		MIC29750	
LX8384		MIC29500	
LX8385		MIC29300	
LX8386		MIC29150	
LX8415		MIC5209	
LX8554		MIC2950X	
LX8584		MIC29502	
LX8585		MIC29510	
LX8586		MIC29710	
LX8587		MIC29300	
LX8610		MIC39100/1/2	
LX8630		MIC2930X	
LX8940	MIC29152		
LX8941		MIC2941	
SG2626	MIC4423		
SG2644	MIC4424		
SG3626	MIC4423		
SG3644	MIC4424		
SG3842	MIC38C42/MIC38HC42		Improved alternative
SG3843	MIC38C43/MIC38HC43		Improved alternative
SG3844	MIC38C44/MIC38HC44		Improved alternative
SG3845	MIC38C45/MIC38HC45		Improved alternative

Maxim

Part Number	Direct Cross	Functional Cross	Comments
ICL7667	MIC4426		
MAX603		MIC5209	
MAX604		MIC5209	
MAX613		MIC2558	
MAX614		MIC2557	
MAX626	MIC4426		
MAX627	MIC4427		
MAX628	MIC4428		
MAX705	MIC705		
MAX706/P/R/S/T	MIC706/P/R/S/T		
MAX707	MIC707		
MAX708/R/S/T	MIC708/R/S/T		
MAX727		MIC4576	
MAX728		MIC4576	
MAX729		MIC4576	
MAX797		MIC2182	
MAX809	MIC809		
MAX810	MIC810		
MAX811	MIC811		
MAX812	MIC812		
MAX825	MIC2775		
MAX869		MIC2545A	
MAX882		MIC5201	
MAX883		MIC5201	
MAX884		MIC5201	

Maxim (continued)

Part Number	Direct Cross	Functional Cross	Comments
MAX890		MIC2544	
MAX891		MIC2544	
MAX892		MIC2544	
MAX893		MIC2544	
MAX894		MIC2546	
MAX895		MIC2546	
MAX1232	MIC1232		
MAX1600		MIC2563	Improved alternative
MAX1602		MIC2562	Improved alternative
MAX1607		MIC2025	Improved alternative
MAX1659		MIC2920	
MAX1693		MIC2025	
MAX1694		MIC2025	
MAX4105		MIC913	
MAX4212		MIC913	
MAX4305		MIC913	
MAX4420	MIC4420		
MAX4426	MIC4426		
MAX4427	MIC4427		
MAX4428	MIC4428		
MAX4429	MIC4429		
MAX6306	MIC2774N		Minor spec. differences, see data sheets
MAX6309	MIC2774L		Minor spec. differences, see data sheets
MAX6312	MIC2774H		Minor spec. differences, see data sheets
MAX6315	MIC6315		
MAX6319	MIC2775		
MAX8863		MIC5245	
MAX8864		MIC5245	
MAX8865		MIC5210	
MAX8866		MIC5210	
MAX8867		MIC5207	
MAX8868		MIC5207	
MAX8869		MIC39100/1/2	
MAX8873		MIC5207	
MAX8874		MIC5207	
MAX8875	MIC5246		
MAX8877	MIC5245		
MAX8878	MIC5245		
MAX8885	MIC5246		
OPA2631		MIC911	
OPA2634		MIC910	
OPA2652		MIC912	
OPA631/2		MIC911	
OPA634/5		MIC910	
TSC426	MIC4426		
TSC427	MIC4427		

Microchip (see also TelCom)

Part Number	Direct Cross	Functional Cross	Comments
MCP100	MIC1815		Minor spec. differences, see data sheets
MCP101	MIC1810		Minor spec. differences, see data sheets
MCP809	MIC809		Minor spec. differences, see data sheets
MCP810	MIC810		Minor spec. differences, see data sheets

Mitel

Part Number	Direct Cross	Functional Cross	Comments
KESRXxxx		MICRF002	

National

Part Number	Direct Cross	Functional Cross	Comments
DS0026		MIC4426	
LM809	MIC809		
LM810	MIC810		
LM2574	LM2574		
LM2575	LM2575		
LM2576	LM2576		
LM2594		MIC4574	
LM2595		MIC4575	
LM2596		MIC4576	
LM2930		MIC2954	
LM2931		MIC2950/MIC2951	
LM2936		MIC5236	
LM2937		MIC5237	
LM2940	MIC2940		Improved alternative
LM2941	MIC2941		Improved alternative
LM385		LM4041	
LM3940	MIC2940A-3.3		Improved alternative
LM4040	LM4040		
LM4041	LM4041		
LM6161		MIC912/MIC911	
LM6171		MIC913/MIC911	
LM7121		MIC913/MIC912	
LM7171		MIC913/MIC912	
LM555		MIC1555	
LMC555		MIC1555	
LMC7101	LMC7101		
LMC7111	LMC7111		
LMC7211	LMC7211		
LMV321		MIC6211	
LP2950	LP2950		
LP2951	LP2951		
LP2954	MIC2954		
LP2980	MIC5203		
LP2980A	MIC5205		Improved alternative
LP2981	MIC5207		
LP2981A	MIC5205		
LP2982	MIC5207		
LP2982	MIC5205		
LP2982A	MIC5205		
LP2985	MIC5245		
LP3962/5		MIC39150	
LP3961/4		MIC39100	
LP3963/6		MIC39300	
MM5450	MM5450		
MM5451	MM5451		
NHM0026	MIC4426		

ON Semiconductor (formerly Motorola)

Part Number	Direct Cross	Functional Cross	Comments
CS-2842	MIC38C42/MIC38HC42		Improved alternative
CS-2843	MIC38C43/MIC38HC43		Improved alternative
CS-3842	MIC38C42/MIC38HC42		Improved alternative
CS-3843	MIC38C43/MIC38HC43		Improved alternative
CS-3972		MIC2172	
CS-5201		MIC2940	
CS-5203		MIC29302	
CS-5204		MIC29502	Higher output current
CS-5205		MIC29502	
CS-5206		MIC29752	Higher output current
CS-5207		MIC29752	
CS-5215		MIC29150	
LM2575	LM2575		
LM2931		MIC2950/MIC2951	
LP2950	LP2950		
LP2951	LP2951		
MAX809	MIC809		
MAX810	MIC810		
MAX811	MIC811		
MAX812	MIC812		
MC33264		MIC5207	
MC33267		MIC5209	
MC33269		MIC2937	
MC78FC		MIC5207	
MC78LC		MIC5203	
MH0026		MIC4426	
NCP1232	MIC1232		
UC2842	MIC38C42/MIC38HC42		Improved alternative
UC2843	MIC38C43/MIC38HC43		Improved alternative
UC2844	MIC38C44/MIC38HC44		Improved alternative
UC2845	MIC38C45/MIC38HC45		Improved alternative
UC3842	MIC38C42/MIC38HC42		Improved alternative
UC3843	MIC38C43/MIC38HC43		Improved alternative
UC3844	MIC38C44/MIC38HC44		Improved alternative
UC3845	MIC38C45/MIC38HC45		Improved alternative

Philips

Part Number	Direct Cross	Functional Cross	Comments
UC2842	MIC38C42/MIC38HC42		Improved alternative
UC2843	MIC38C43/MIC38HC43		Improved alternative
UC2844	MIC38C44/MIC38HC44		Improved alternative
UC2845	MIC38C45/MIC38HC45		Improved alternative
UC78PC33		MIC5245	

Ricoh

Part Number	Direct Cross	Functional Cross	Comments
R1110N		MIC5207	
R1120N		MIC5207	

Samsung

Part Number	Direct Cross	Functional Cross	Comments
KA3842	MIC38C42/MIC38HC42		Improved alternative
KA3843	MIC38C43/MIC38HC43		Improved alternative
KA3844	MIC38C44/MIC38HC44		Improved alternative
KA3845	MIC38C45/MIC38HC45		Improved alternative

Semtech

Part Number	Direct Cross	Functional Cross	Comments
EC2950	MIC2950		
EC2951	MIC2951		
EZ1083		MIC29750	
EZ1084		MIC29500	
EZ1085		MIC29300	
EZ1086		MIC29150	
EZ1117		MIC2937	
EZ1584		MIC29710	
EZ1585		MIC29510	
EZ1587		MIC29310	
EZ55Z3L		MIC2954	
LM2575	LM2575		
LM2576	LM2576		
LP2951	LP2951		
SC1565	MIC39150		
SC1566	MIC39300		
SC2980	MIC5205		
SC5203	MIC5203		
SC5205	MIC5205		
SC5825	MIC2525		
SC5826	MIC2526		

Seiko

Part Number	Direct Cross	Functional Cross	Comments
S80230	MIC5203		
S80250	MIC5203		
S81215	MIC5203		
S81230	MIC5203		
S81233	MIC5203		
S81240	MIC5203		
S81250	MIC5203		
S81330	MIC5207		
S81340	MIC5207		
S81350	MIC5207		

SGS-Thomson

Part Number	Direct Cross	Functional Cross	Comments
LF120	MIC5209		
LF33	MIC5209		
LF50	MIC5209		
M5450	MM5450		
M5451	MM5450		
SGS2626	MIC4423		
SGS3626	MIC4423		

SGS-Thomson (continued)

Part Number	Direct Cross	Functional Cross	Comments
UC3842	MIC38C42/MIC38HC42		Improved alternative
UC3843	MIC38C43/MIC38HC43		Improved alternative
UC3844	MIC38C44/MIC38HC44		Improved alternative
UC3845	MIC38C45/MIC38HC45		Improved alternative

Siliconix (Vishay)

Part Number	Direct Cross	Functional Cross	Comments
SG2626	MIC4423		
SG3626	MIC4423		
SI9400		MIC94001	

Telcom (now part of Microchip)

Part Number	Direct Cross	Functional Cross	Comments
TC1014		MIC5207	
TC1015		MIC5207	
TC1054		MIC5206	
TC1055		MIC5206	
TC1070		MIC5207	
TC1071		MIC5207	
TC1072		MIC5206	
TC1073		MIC5206	
TC1185	MIC5245		
TC1186		MIC5206	
TC1187		MIC5207	
TC1232	MIC1232		
TC1270	MIC811		
TC1271	MIC812		
TC1272	MIC1810		
TC1275	MIC1815		
TC1410		MIC4416	
TC1411		MIC4416	
TC1413	MIC4429		Improved alternative
TC1426	MIC4426		
TC1427	MIC4427		
TC1428	MIC4428		
TC18C43	MIC38C43/MIC38HC43		
TC18C46	MIC38C43/MIC38HC43		
TC28C43	MIC38C43/MIC38HC43		
TC28C46	MIC38C43/MIC38HC43		
TC38C43	MIC38C43/MIC38HC43		
TC38C46	MIC38C43/MIC38HC43		
TC426	MIC4426		
TC427	MIC4427		
TC428	MIC4428		
TC4420	MIC4420		
TC4421	MIC4421		
TC4422	MIC4422		
TC4423	MIC4423		
TC4424	MIC4424		
TC4425	MIC4425		
TC4426	MIC4426		
TC4427	MIC4427		

Telcom (continued)

Part Number	Direct Cross	Functional Cross	Comments
TC4428	MIC4428		
TC4429	MIC4429		
TC4431		MIC4429	
TC4432		MIC4420	
TC4467	MIC4467		
TC4468	MIC4468		
TC4469	MIC4469		
TC55		MIC2954	
TCL1117		MIC2937	
TCL1580		MIC29712	
TCL1584		MIC29710	
TCL1585		MIC29510	
TCL1587		MIC29310	
TCM809	MIC809		
TCM810	MIC810		
TCN4040	LM4040		
TCN4041	LM4041		

Texas Instruments (Unitrode)

Part Number	Direct Cross	Functional Cross	Comments
LM2576	LM2576		
LT751M	MIC29152		
LT1072		MIC2172	
SN75518	MIC5818		
TL4810	MIC5810		
TL750L		MIC2950	
TL750M	MIC2937		
TL751L		MIC2951	
TL75LP		MIC2920	
THS4001		MIC912/MIC913	
THS4031/2		MIC912	
THS4041/2		MIC912	
THS4051/2		MIC911	
THS4061/2		MIC911	
THS4081/2		MIC911	
TPS2041/2051		MIC2025	
TPS2042/2052		MIC2026	
TPS2044/2054		MIC2027	
TPS2014		MIC2025-2	
TPS2015		MIC2025-2	
TPS2201		MIC2563A	
TPS2205		MIC2563A	
TPS2206		MIC2564A	
TPS2211		MIC2562A	
TPS2214		MIC2564A	
TPS2216		MIC2564A	
TPS2811		MIC4426	
TPS2812		MIC4427	
TPS2813		MIC4428	
TPS3705		MIC706	
TPS3707		MIC708	
TPS3809		MIC809	Minor spec differences, see data sheets

Texas Instruments (continued)

Part Number	Direct Cross	Functional Cross	Comments
TPS3825		MIC2775	Minor spec differences, see data sheets
TPS7101		MIC5209/MIC5216	
TPS7133		MIC5209/MIC5216	
TPS7148		MIC5209/MIC5216	
TPS7150		MIC5209/MIC5216	
TPS7201		MIC5209/MIC5216	
TPS7233		MIC5209/MIC5216	
TPS7248		MIC5209/MIC5216	
TPS7250		MIC5209/MIC5216	
UC2575	LM2575		
UC2576	LM2576		
UC2842	MIC38C42/MIC38HC42		Improved alternative
UC2843	MIC38C43/MIC38HC43		Improved alternative
UC2844	MIC38C44/MIC38HC44		Improved alternative
UC2845	MIC38C45/MIC38HC45		Improved alternative
UC3842	MIC38C42/MIC38HC42		Improved alternative
UC3843	MIC38C43/MIC38HC43		Improved alternative
UC3844	MIC38C44/MIC38HC44		Improved alternative
UCC2800		MIC38C43/MIC38HC43	Smaller package available
UCC2801		MIC38C45/MIC38HC45	
UCC2802		MIC38C42/MIC38HC42	
UCC2804		MIC38C44/MIC38HC44	
UCC3800		MIC38C43/MIC38HC43	Smaller package available
UCC3801		MIC38C45/MIC38HC45	
UCC3802		MIC38C42/MIC38HC42	
UCC3804		MIC38C44/MIC38HC44	

Toko

Part Number	Direct Cross	Functional Cross	Comments
TK112xx		MIC5207/MIC5203	
TK113xx		MIC5207/MIC5203	
TK116xx		MIC5207/MIC5203	
TK119xx		MIC5207/MIC5203	
TK711xx	MIC2950		
TK712xx		MIC5207/MIC5203	
TK713xx		MIC5203	

Zetek

Part Number	Direct Cross	Functional Cross	Comments
ZXCM209	MIC810		
ZXCM210	MIC810		

Notes:

Direct cross parts are parametrically similar and are available in similar packages.

Functional cross parts are functionally similar, but have different specifications, and may have different packages.

High Bandwidth Cross Reference

Fairchild Semiconductor

Part Number	Direct Cross	Functional Cross	Package
<i>Super 300K™ Logic</i>			
FSC100301	SY100S301		FC, JC, JCTR
FSC100302	SY100S302		FC, JC, JCTR
FSC100304	SY100S304		FC, JC, JCTR
FSC100307	SY100S307		FC, JC, JCTR
FSC100313	SY100S313		FC, JC, JCTR
FSC100314	SY100S314		FC, JC, JCTR
FSC100317	SY100S317		FC, JC, JCTR
FSC100318	SY100S318		FC, JC, JCTR
FSC100321	SY100S321		FC, JC, JCTR
FSC100322	SY100S322		FC, JC, JCTR
FSC100324	SY100S324		FC, JC, JCTR
FSC100325	SY100S325		FC, JC, JCTR
FSC100331	SY100S331		FC, JC, JCTR
FSC100336	SY100S336		FC, JC, JCTR
FSC100341	SY100S341		FC, JC, JCTR
FSC100350	SY100S350		FC, JC, JCTR
FSC100351	SY100S351		FC, JC, JCTR
FSC100355	SY100S355		FC, JC, JCTR
FSC100360	SY100S360		FC, JC, JCTR
FSC100363	SY100S363		FC, JC, JCTR
FSC100364	SY100S364		FC, JC, JCTR
FSC100366	SY100S366		FC, JC, JCTR
FSC100370	SY100S370		FC, JC, JCTR
FSC100371	SY100S371		FC, JC, JCTR
FSC100391	SY100S391		FC, JC, JCTR

ON Semiconductor Index

Part Number	Direct Cross	Functional Cross	Package
MC100E016	SY100E016		JC, JCTR
MC100E101	SY100E101		JC, JCTR
MC100E104	SY100E104		JC, JCTR
MC100E107	SY100E107		JC, JCTR
MC100E111	SY100E111		JC, JCTR
MC100LVE111	SY100E111L		JC, JCTR
MC100E112	SY100E112		JC, JCTR
MC100E116	SY100E116		JC, JCTR
MC100E122	SY100E122		JC, JCTR
MC100E131	SY100E131		JC, JCTR
MC100E136	SY100E136		JC, JCTR
MC100E137	SY100E137		JC, JCTR
MC100E141	SY100E141		JC, JCTR
MC100E142	SY100E142		JC, JCTR
MC100E143	SY100E143		JC, JCTR
MC100E150	SY100E150		JC, JCTR
MC100E151	SY100E151		JC, JCTR
MC100E154	SY100E154		JC, JCTR
MC100E155	SY100E155		JC, JCTR
MC100E156	SY100E156		JC, JCTR
MC100E157	SY100E157		JC, JCTR
MC100E158	SY100E158		JC, JCTR
MC100E160	SY100E160		JC, JCTR
MC100E163	SY100E163		JC, JCTR
MC100E164	SY100E164		JC, JCTR
MC100E166	SY100E166		JC, JCTR
MC100E167	SY100E167		JC, JCTR
MC100E171	SY100E171		JC, JCTR
MC100E175	SY100E175		JC, JCTR
MC100E193	SY100E193		JC, JCTR
MC100E195	SY100E195		JC, JCTR
MC100E196	SY100E196		JC, JCTR
MC100E212	SY100E212		JC, JCTR
MC100E241	SY100E241		JC, JCTR
MC100E256	SY100E256		JC, JCTR
MC100E336	SY100E336		JC, JCTR
MC100E337	SY100E337		JC, JCTR
MC100E404	SY100E404		JC, JCTR
MC100E416	SY100E416		JC, JCTR
MC100E417	SY100E417		JC, JCTR
MC100E431	SY100E431		JC, JCTR
MC100E445	SY100E445		JC, JCTR
MC100E446	SY100E446		JC, JCTR
MC100E451	SY100E451		JC, JCTR
MC100E452	SY100E452		JC, JCTR
MC100E457	SY100E457		JC, JCTR
MC100EL01	SY100EL01		ZC, ZCTR
MC100EL04	SY100EL04		ZC, ZCTR
MC100EL05	SY100EL05		ZC, ZCTR
MC100EL07	SY100EL07		ZC, ZCTR
MC100EL11	SY100EL11V		ZC, ZCTR

Part Number	Direct Cross	Functional Cross	Package
MC100LVEL11	SY100EL11V		ZC, ZCTR
MC100EL12	SY100EL12		ZC, ZCTR
MC100EL14	SY100EL14V		ZC, ZCTR
MC100LVEL14	SY100EL14V		ZC, ZCTR
MC100EL15	SY100EL15		ZC, ZCTR
MC100EL16	SY100EL16V		KC, KCTR, ZC, ZCTR
MC100LVEL16	SY100EL16V		KC, KCTR, ZC, ZCTR
MC100EL17	SY100EL17V		ZC, ZCTR
MC100LVEL17	SY100EL17V		ZC, ZCTR
MC100EL29	SY100EL29V		ZC, ZCTR
MC100LVEL29	SY100EL29V		ZC, ZCTR
MC100EL31	SY100EL31		ZC, ZCTR
MC100EL32	SY100EL32V		ZC, ZCTR
MC100LVEL32	SY100EL32V		ZC, ZCTR
MC100EL33	SY100EL33		ZC, ZCTR
MC100LVEL33	SY100EL33L		ZC, ZCTR
MC100EL34	SY100EL34		ZC, ZCTR
MC100EL35	SY100EL35		ZC, ZCTR
MC100EL38	SY100EL38		ZC, ZCTR
MC100LVEL38	SY100EL38L		ZC, ZCTR
MC100EL39	SY100S839V		ZC, ZCTR
MC100LVEL39	SY100S839V		ZC, ZCTR
MC100EL51	SY100EL51		ZC, ZCTR
MC100EL52	SY100EL52		ZC, ZCTR
MC100EL56	SY100EL56V		ZC, ZCTR
MC100LVEL56	SY100EL56V		ZC, ZCTR
MC100EL57	SY100EL57		ZC, ZCTR
MC100EL58	SY100EL58		ZC, ZCTR
MC100EL90	SY100EL90V		ZC, ZCTR
MC100LVEL90	SY100EL90V		ZC, ZCTR
MC100EL91	SY100EL91		ZC, ZCTR
MC100LVEL91	SY100EL91L		ZC, ZCTR
MC100EL92	SY100EL92		ZC, ZCTR
MC100LVEL92	SY100EL92		ZC, ZCTR
MC100ELT20	SY100ELT20V		ZC, ZCTR
MC100ELT21	SY100ELT21		ZC, ZCTR
MC100ELT22	SY100ELT22		ZC, ZCTR
MC100LVELT22	SY100ELT22L		ZC, ZCTR
MC100ELT23	SY100ELT23		ZC, ZCTR
MC100LVELT23	SY100ELT23L		ZC, ZCTR
MC100ELT24	SY100ELT24		ZC, ZCTR
MC100ELT25	SY100ELT25		ZC, ZCTR
MC10EP01	SY10EP01V		KC, KCTR, ZC, ZCTR
MC10EP16	SY10EP16V		KC, KCTR, ZC, ZCTR
MC10EP58	SY10EP58V		KC, KCTR, ZC, ZCTR
MC100H600	SY100H600		JC, JCTR
MC100H601	SY100H601		JC, JCTR
MC100H602	SY100H602		JC, JCTR
MC100H603	SY100H603		JC, JCTR
MC100H606	SY100H606		JC, JCTR
MC100H607	SY100H607		JC, JCTR
MC100H641	SY100H641		JC, JCTR

Part Number	Direct Cross	Functional Cross	Package
MC100H646	SY100H646L		JC, JCTR
MC100H643	SY100HA643		JC, JCTR
MC10E016	SY10E016		JC, JCTR
MC10E101	SY10E101		JC, JCTR
MC10E104	SY10E104		JC, JCTR
MC10E107	SY10E107		JC, JCTR
MC10E111	SY10E111		JC, JCTR
MC10E112	SY10E112		JC, JCTR
MC10E116	SY10E116		JC, JCTR
MC10E122	SY10E122		JC, JCTR
MC10E131	SY10E131		JC, JCTR
MC10E136	SY10E136		JC, JCTR
MC10E137	SY10E137		JC, JCTR
MC10E141	SY10E141		JC, JCTR
MC10E142	SY10E142		JC, JCTR
MC10E143	SY10E143		JC, JCTR
MC10E150	SY10E150		JC, JCTR
MC10E151	SY10E151		JC, JCTR
MC10E154	SY10E154		JC, JCTR
MC10E155	SY10E155		JC, JCTR
MC10E156	SY10E156		JC, JCTR
MC10E157	SY10E157		JC, JCTR
MC10E158	SY10E158		JC, JCTR
MC10E160	SY10E160		JC, JCTR
MC10E163	SY10E163		JC, JCTR
MC10E164	SY10E164		JC, JCTR
MC10E166	SY10E166		JC, JCTR
MC10E167	SY10E167		JC, JCTR
MC10E171	SY10E171		JC, JCTR
MC10E175	SY10E175		JC, JCTR
MC10E193	SY10E193		JC, JCTR
MC10E195	SY10E195		JC, JCTR
MC10E196	SY10E196		JC, JCTR
MC10E212	SY10E212		JC, JCTR
MC10E241	SY10E241		JC, JCTR
MC10E256	SY10E256		JC, JCTR
MC10E336	SY10E336		JC, JCTR
MC10E337	SY10E337		JC, JCTR
MC10E404	SY10E404		JC, JCTR
MC10E416	SY10E416		JC, JCTR
MC10E431	SY10E431		JC, JCTR
MC10E445	SY10E445		JC, JCTR
MC10E446	SY10E446		JC, JCTR
MC10E451	SY10E451		JC, JCTR
MC10E452	SY10E452		JC, JCTR
MC10E457	SY10E457		JC, JCTR
MC10EL01	SY10EL01		ZC, ZCTR
MC10EL04	SY10EL04		ZC, ZCTR
MC10EL05	SY10EL05		ZC, ZCTR
MC10EL07	SY10EL07		ZC, ZCTR
MC10EL11	SY10EL11V		ZC, ZCTR
MC10EL12	SY10EL12		ZC, ZCTR
MC10EL15	SY10EL15		ZC, ZCTR

Part Number	Direct Cross	Functional Cross	Package
MC10EL16	SY10EL16V		KC, KCTR, ZC, ZCTR
MC10EL31	SY10EL31		ZC, ZCTR
MC10EL32	SY10EL32V		ZC, ZCTR
MC10EL33	SY10EL33		ZC, ZCTR
MC10EL34	SY10EL34		ZC, ZCTR
MC10EL35	SY10EL35		ZC, ZCTR
MC10EL38	SY10EL38		ZC, ZCTR
MC10EL51	SY10EL51		ZC, ZCTR
MC10EL52	SY10EL52		ZC, ZCTR
MC10EL57	SY10EL57		ZC, ZCTR
MC10EL58	SY10EL58		ZC, ZCTR
MC10EL89	SY10EL89		ZC, ZCTR
MC10SX1189	SY10EL1189		ZC, ZCTR
MC10ELT20	SY10ELT20V		ZC, ZCTR
MC10ELT21	SY10ELT21		ZC, ZCTR
MC10ELT22	SY10ELT22		ZC, ZCTR
MC10ELT23	SY10ELT23		ZC, ZCTR
MC10EP01	SY10EP01V		KC, KCTR, ZC, ZCTR
MC10EP16	SY10EP16V		KC, KCTR, ZC, ZCTR
MC10EP58	SY10EP58V		KC, KCTR, ZC, ZCTR
MC10H350	SY10H350		JC, JCTR
MC10H351	SY10H351		JC, JCTR
MC10H352	SY10H352		JC, JCTR
MC10H600	SY10H600		JC, JCTR
MC10H601	SY10H601		JC, JCTR
MC10H602	SY10H602		JC, JCTR
MC10H603	SY10H603		JC, JCTR
MC10H606	SY10H606		JC, JCTR
MC10H607	SY10H607		JC, JCTR
MC10H641	SY10H641		JC, JCTR
MC10H646	SY10H646L		JC, JCTR
MC12429		SY89429A/V*	ZC, ZCTR, JC, JCTR
MC12439		SY89430V*	ZC, ZCTR, JC, JCTR

* For crosses, refer to the following application notes at www.micrel.com:

SY89429A Frequency Synthesis

SY89429/30V Frequency Synthesis

How to use SY89430V in Other Applications

ON

Semiconductor

Micrel High Bandwidth Package Codes

FN	JC	PLCC, commercial temperature range
FN, FNR2	JCTR:	PLCC, commercial temperature range, tape and reel
D, DW	ZC	SOIC, commercial temperature range
DR2, DWR2	ZCTR	SOIC, commercial temperature range, tape and reel
DT	KC	MSOP, commercial temperature range
DTR2	KCTR	MSOP, commercial temperature range, tape and reel

Fairchild

Micrel High Bandwidth Package Codes

	FC:	flatpack, commercial temperature range (Fairchild only)
QC	JC	PLCC, commercial temperature range
QCX	JCTR:	PLCC, commercial temperature range, tape and reel

Packaging for Automatic Handling—Tape & Reel and Ammo Pack

General Description

Tape & Reel

Surface mount and TO-92 devices are available in tape and reel packaging. Surface mount components are retained in an embossed carrier tape by a cover tape. TO-92 device leads are secured to a backing tape by a cover tape. The tape is spooled on standard size reels.

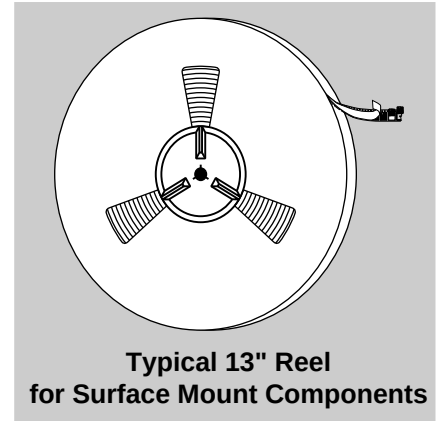
Ammo Pack

TO-92 devices are also available in an “ammo pack.” TO-92 devices are secured to a backing tape by a cover tape and are fanfolded into a box. Ammo packs contain the same quantity, feed direction, and component orientation as a reel.

To order, specify the complete part number with the suffix “AP” (*example[†]: MICxxxxZ AP*).

Pricing

Contact the factory for price adder and availability.



Tape & Reel Standards

Embossed tape and reel packaging conforms to:

- 8mm & 12mm Taping of Surface Mount Components for Automatic Handling, EIA-481-1*
- 16mm and 24mm Embossed Carrier Taping of Surface Mount Components for Automatic Handling, EIA-481-2*
- 32mm, 44mm and 56mm Embossed Carrier Taping of Surface Mount Components for Automatic Handling, EIA-481-3*

Packages Available in Tape & Reel

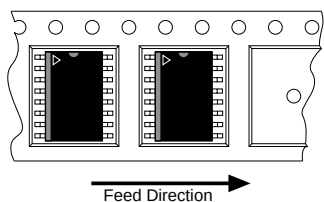
Part Number [†]	Package Description	Quantity / Reel	Reel Diameter	Carrier Tape Width	Carrier Tape Pitch
MICxxxxM T&R	SOP-8	2,500	13"	12mm	8mm
	SOP-14	2,500	13"	16mm	8mm
	SOP-16	2,500	13"	16mm	8mm
MICxxxxWM T&R	Wide SOP-16	1,000	13"	16mm	12mm
	Wide SOP-18	1,000	13"	16mm	12mm
	Wide SOP-20	1,000	13"	24mm	12mm
	Wide SOP-24	1,000	13"	24mm	12mm
MICxxxxMM T&R	MM8™ MSOP-8	2,500	13"	12mm	8mm
MICxxxxSM T&R	SSOP-28	2,000	13"	16mm	12mm
MICxxxxTS T&R	TSSOP-14	2,500	13"	12mm	8mm
MICxxxxV T&R	PLCC-20	1,000	13"	16mm	12mm
	PLCC-28	500	13"	24mm	16mm
	PLCC-44	500	13"	32mm	24mm
MICxxxxC5 T&R	SC-70	3,000	7"	8mm	4mm
MICxxxxM4 T&R	SOT-143	3,000	7"	8mm	4mm
MICxxxxM3 T&R	SOT-23	3,000	7"	8mm	4mm
MICxxxxM5 T&R	SOT-23-5	3,000	7"	8mm	4mm
MICxxxxM6 T&R	SOT-23-6	3,000	7"	8mm	4mm
MICxxxxS T&R	SOT-223	2,500	13"	16mm	12mm
MICxxxxU T&R	TO-263[-3]	750	13"	24mm	16mm
	TO-263-5	750	13"	24mm	16mm
MICxxxxZ T&R	TO-92	2,000	14¼"‡	—	1/2"

* Standards are available from: Electronic Industries Associations, EIA Standards Sales Department, tel: (202) 457-4966

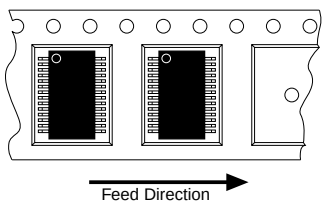
[†] xxxx = base part number + temperature designation. Example: MIC2557BM T&R

[‡] Cardboard reel

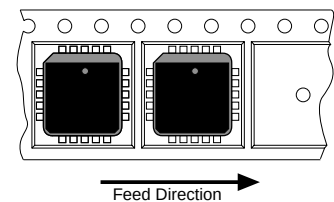
Package Orientation



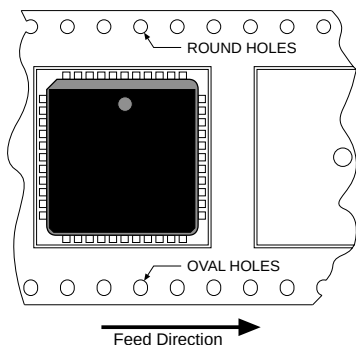
Typical SOIC Package Orientation
12mm, 16mm, 24mm Carrier Tape



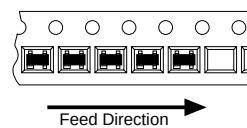
Typical SSOP Package Orientation
16mm Carrier Tape



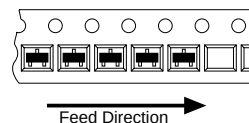
Typical PLCC Package Orientation
16mm, 24mm Carrier Tape



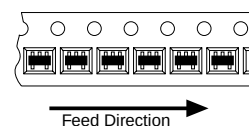
Typical PLCC Package Orientation
32mm Carrier Tape



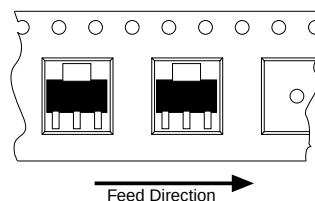
SOT-143 Package Orientation
8mm Carrier Tape



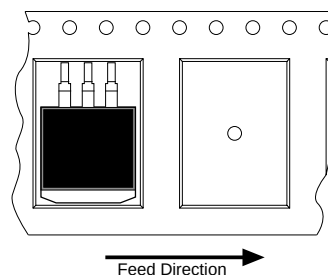
SOT-23 Package Orientation
8mm Carrier Tape



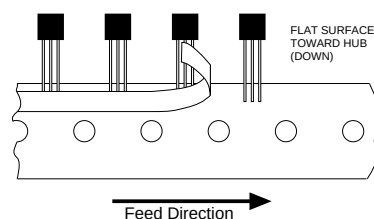
SC-70, SOT-23-5 Package Orientation
8mm Carrier Tape



SOT-223 Package Orientation
16mm Carrier Tape



Typical TO-263 Package Orientation
24mm Carrier Tape



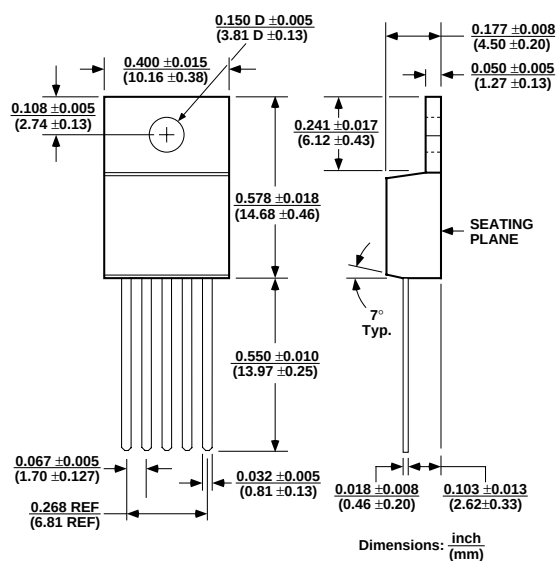
Typical TO-92 Package Orientation

Lead Bend Options—5-Lead TO-220 Package

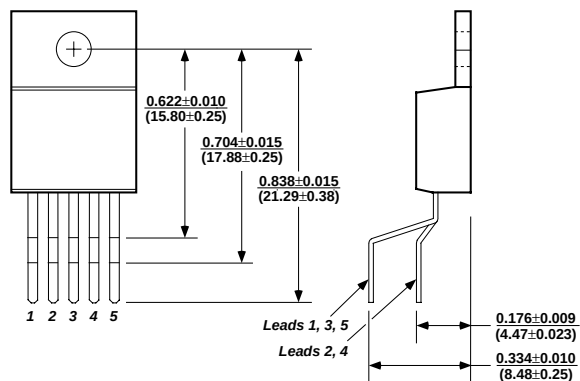
Part Number	Package	Replaces National Semi.
MICxxxxxyT-LB02	5-lead TO-220	TA05B (-LB02), T05C*
MICxxxxxyT-LB03	5-lead TO-220	T05D (-LB-03), T05B*

* Leads not trimmed after bending.

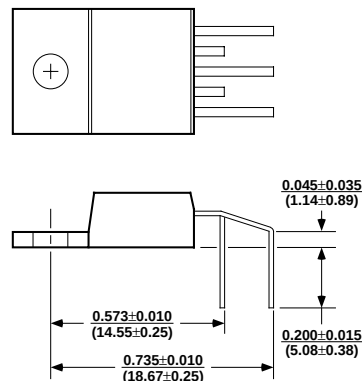
*1000-piece minimum order.
Contact factory for price adder and availability.*



5-lead TO-220 Basic Dimensions

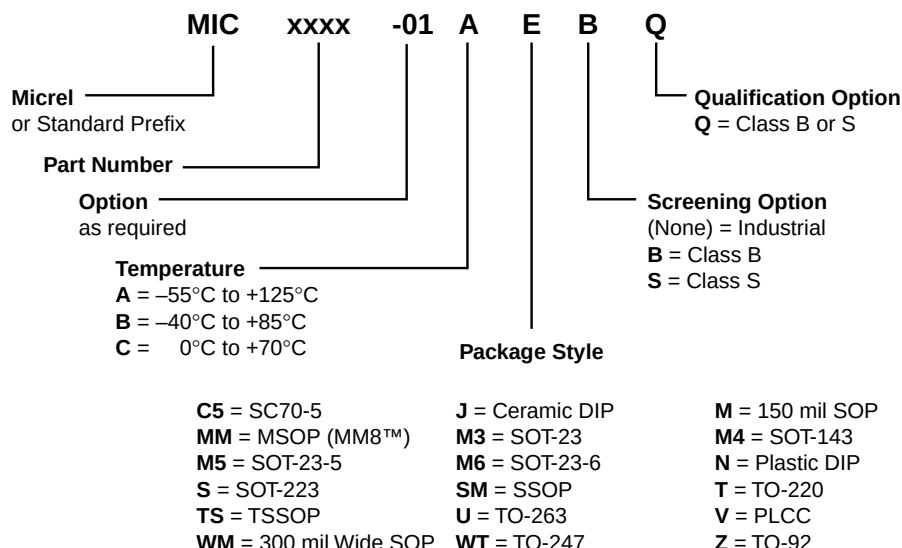


TO-220 Vertical Lead Bend Option -LB03



TO-220 Horizontal Lead Bend Option -LB02

Analog Part Identification



Micrel Die is distributed by:

NORTH AMERICA — DIE DISTRIBUTION ONLY

Chip Supply, Inc.

7225 Orange Blossom Trail
 Orlando, FL
 USA 32810-2696

Tel: + 1 (407) 298-7100
 Fax: + 1 (407) 290-0164

EUROPE — DIE DISTRIBUTION ONLY

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5 Queen Street
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 Fax: + 44 (1616) 260380

Eltek Semiconductor, Ltd.

Nelson Road Industrial Estate
 Dartmouth
 Devon TQ6 9LA
 United Kingdom

Tel: + 44 (1803) 834455
 Fax: + 44 (1803) 833011

Analog Part Identification

Teeny™, IttyBitty®, and TinyFET® Part Identification

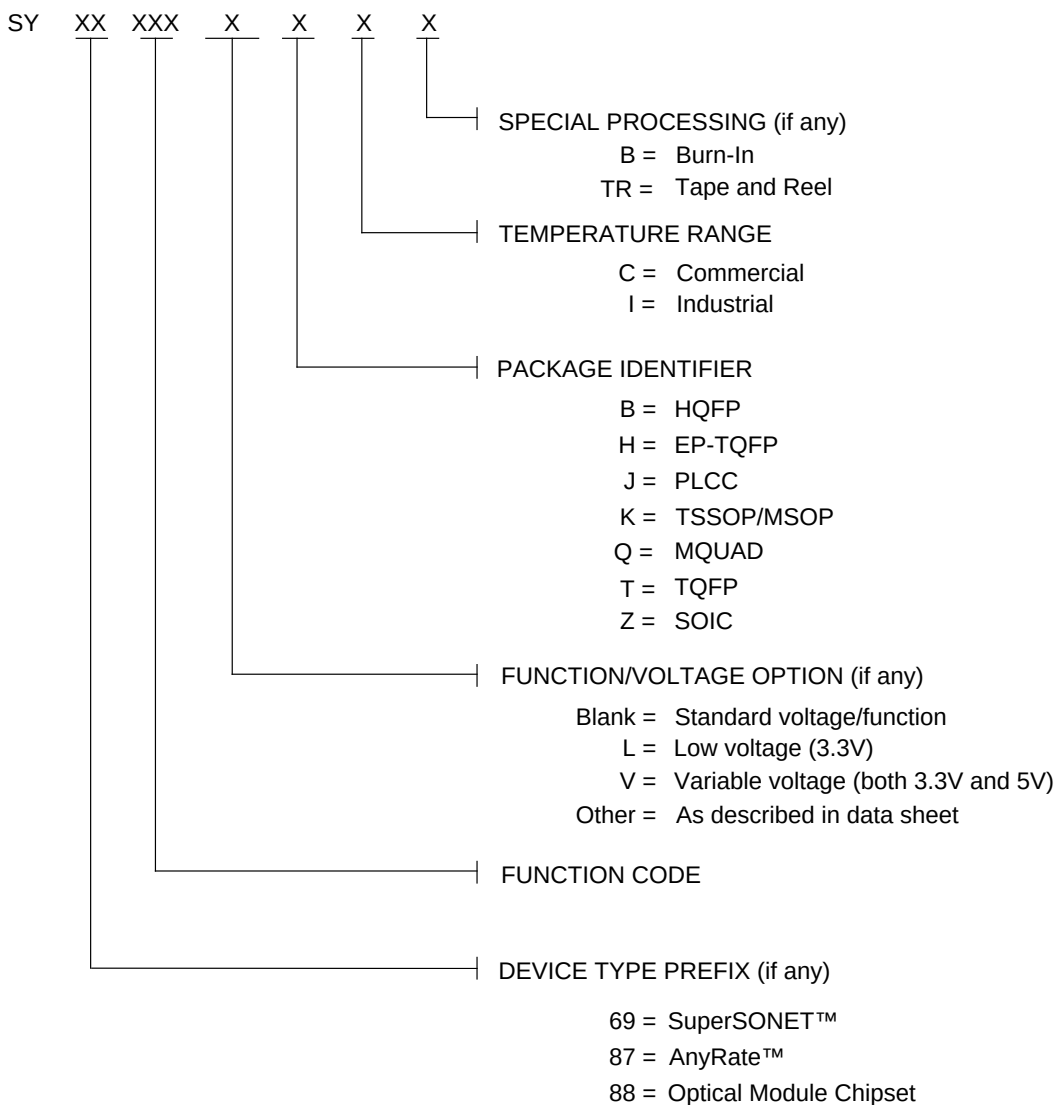
Teeny™ SC-70, IttyBitty®, TinyFET®, and other SOT-143, or SOT-23 packaged devices use abbreviated markings for identification because of their small size.

Mark	Part Number	Description	Note
A11	MIC6211BM5	IttyBitty® Op Amp	
A12	LMC7101BM5	Low-Power Operational Amplifier	
A12A	LMC7101AIM5	[Improved Precision] Low-Power Operational Amplifier	
A13	MIC7111BM5	2.4V IttyBitty® Rail-to-Rail Input/Output Op Amp	
A14	MIC7211BM5	IttyBitty® Rail-to-Rail Input Comparator	push-pull output
A15	MIC7221BM5	IttyBitty® Rail-to-Rail Input Comparator	open-drain output
A16	MIC7201BM5	GainBlock™ Difference Amplifier	
A17	MIC7300BM5	High-Output Drive Rail-to-Rail Op Amp	
A21	MIC910BM5	135MHz Low-Power SOT-23-5 Op Amp	
A22	MIC911BM5	105MHz Low-Power SOT23-5 Op Amp	
A23	MIC912BM5	200MHz Low-Power SOT-23-5 Op Amp	
A24	MIC913BM5	350MHz Low-Power SOT-23-5 Op Amp	
A25	MIC820BC5	5MHz Ultra-Low Power 5V Op Amp	
A26	MIC914BM5	160MHz Low-Power SOT-23-5 Op Amp	
A51	MIC6251BM5	IttyBitty® Instrumentation Amplifier	+2, +1, -1 gain amp.
A52	MIC6252BM5	IttyBitty® Instrumentation Amplifier	+0.5, +1 gain amp., avg. value amp.
B10	MIC6270BM5	IttyBitty® Comparator	
B11	MIC833	Latched Comparator with Reference	
B12	MIC834	Comparator with Reference	
B13	MIC841HBC5	Comparator with Reference in SC70-5	
B14	MIC841LBC5	Comparator with Reference in SC70-5	
B15	MIC841NBC5	Comparator with Reference in SC70-5	
B16	MIC842HBC5	Comparator with Reference in SC70-5	
B17	MIC842LBC5	Comparator with Reference in SC70-5	
B18	MIC842NBC5	Comparator with Reference in SC70-5	
C10	MIC2660BM5	IttyBitty® Charge Pump	
D10	MIC4416BM4	IttyBitty® Low-Side MOSFET Driver	noninverting
D11	MIC4417BM4	IttyBitty® Low-Side MOSFET Driver	inverting
ML10	MIC4416BM4	IttyBitty® Low-Side MOSFET Driver	early production mark
F10	MIC2514BM5	IttyBitty® Integrated High-Side Switch	
H10	MIC5018BM4	IttyBitty® High-Side MOSFET Driver	noninverting
MH10	MIC5018BM4	IttyBitty® High-Side MOSFET Driver	early production mark
IJ	ETC/MIC809JU	Microprocessor Reset Circuit	4.00V
IL	ETC/MIC809LU...	Microprocessor Reset Circuit	4.63V
IM	ETC/MIC809MU...	Microprocessor Reset Circuit	4.38V
IR	ETC/MIC809RU...	Microprocessor Reset Circuit	2.63V
IS	ETC/MIC809SU...	Microprocessor Reset Circuit	2.93V
IT	ETC/MIC809TU...	Microprocessor Reset Circuit	3.08V
JL	ETC/MIC810LU...	Microprocessor Reset Circuit	4.63V
JM	ETC/MIC810MU...	Microprocessor Reset Circuit	4.38V
JR	ETC/MIC810RU...	Microprocessor Reset Circuit	2.63V
JS	ETC/MIC810SU...	Microprocessor Reset Circuit	2.93V
JT	ETC/MIC810TU...	Microprocessor Reset Circuit	3.08V
KJ	ETC/MIC811JU	Microprocessor Reset Circuit	4.00V
KL	ETC/MIC811LU...	Microprocessor Reset Circuit	4.63V
KM	ETC/MIC811MU...	Microprocessor Reset Circuit	4.38V
KR	ETC/MIC811RU...	Microprocessor Reset Circuit	2.63V
KS	ETC/MIC811SU...	Microprocessor Reset Circuit	2.93V
KT	ETC/MIC811TU...	Microprocessor Reset Circuit	3.08V

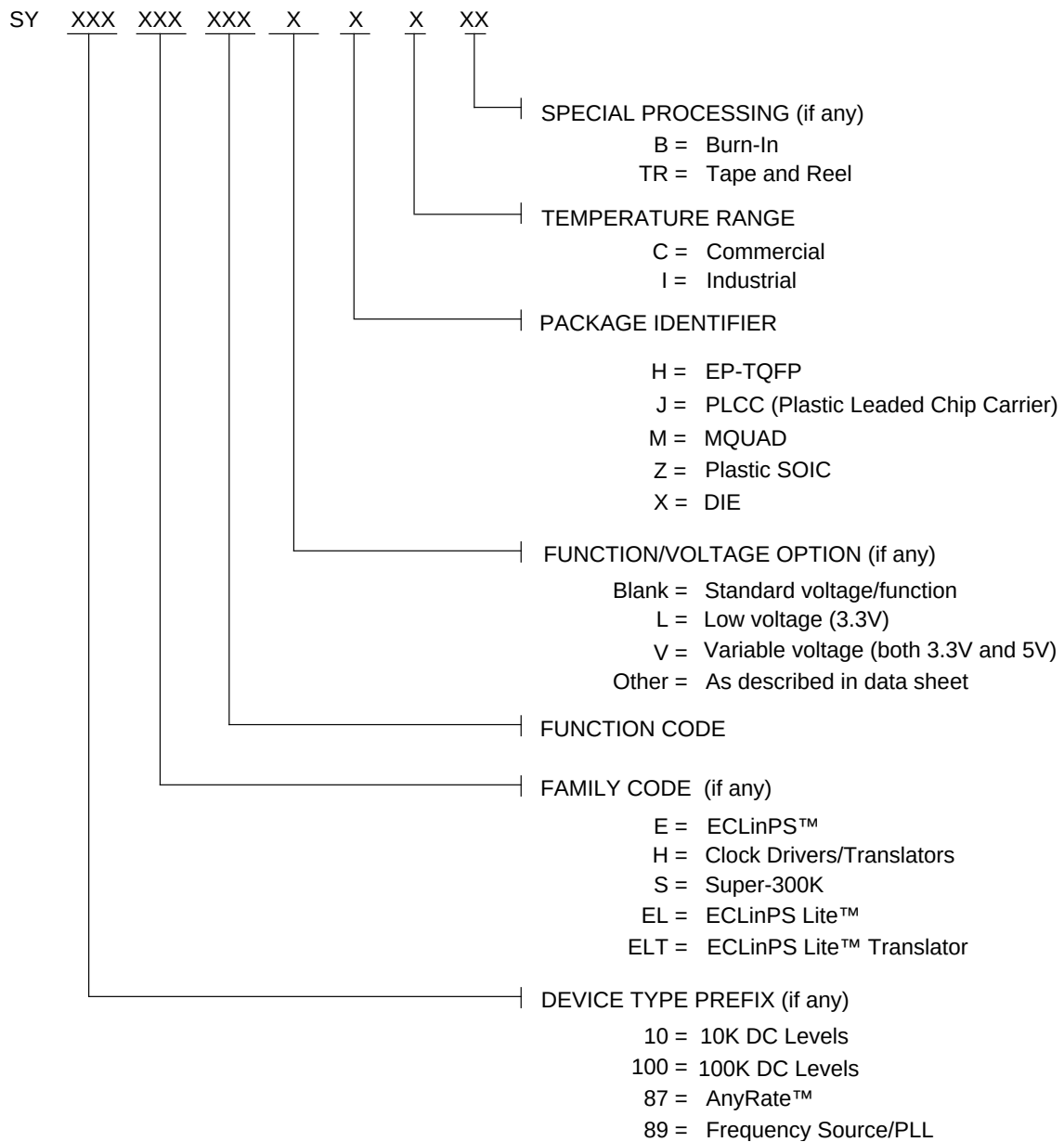
Analog Part Identification

LAx	MIC5213-x.xBC5	Teeny™ SC-70 µCap LDO Regulator	
LAXx	MIC5203-x.xBM4	80mA Low-Dropout Regulator (SOT-143)	fixed output (also see LK)
LBAA	MIC5205BM5	150mA Low-Dropout Regulator	adjustable output
LBxx	MIC5205-x.xBM5	150mA Low-Dropout Regulator	fixed output
LC50	MIC5230-5.0BM5	10mA Microcurrent Voltage Regulator	5.0V output
LDxx	MIC5206-x.xBM5	150mA Low-Noise LDO Regulator	fixed output
LEAA	MIC5207BM5	180mA Low-Noise LDO Regulator	adjustable output
LExx	MIC5207-x.xBM5	180mA Low-Noise LDO Regulator	fixed output
LFxx	MIC5211-x.xBM5	Dual 50mA LDO Voltage Regulator	fixed outputs, special code
LGAA	MIC5219BM5	500mA-Peak Output LDO Regulator	adjustable output
LGxx	MIC5219-x.xBM5	500mA-Peak Output LDO Regulator	fixed output
LHxx	MIC5216-x.xBM5	500mA-Peak Output LDO Regulator	fixed output
LKxx	MIC5203-x.xBM5	80mA Low-Dropout Regulator (SOT-23-5)	fixed output (also see LA)
LL	ETC/MIC812LU...	Microprocessor Reset Circuit	4.63V
LLxx	MIC5270-x.xBM5	IttyBitty® Negative Low Dropout Regulator	fixed output
LM	ETC/MIC812MU...	Microprocessor Reset Circuit	4.38V
LR	ETC/MIC812RU...	Microprocessor Reset Circuit	2.63V
LS	ETC/MIC812SU...	Microprocessor Reset Circuit	2.93V
LSxx	MIC5245-x.xBM5	150mA IttyBitty® CMS LDO Regulator	fixed output
LT	ETC/MIC812TU...	Microprocessor Reset Circuit	3.08V
NA	ETC/MIC1810-5U	Microprocessor Reset Circuit	4.62V
NB	ETC/MIC1810-10U	Microprocessor Reset Circuit	4.37V
NC	ETC/MIC1810-15U	Microprocessor Reset Circuit	4.12V
ND	ETC/MIC1815-10U	Microprocessor Reset Circuit	2.88V
NE	ETC/MIC1815-20U	Microprocessor Reset Circuit	2.55V
NR	ETC/MIC6315-26D3U	Open-Drain µP Reset Circuit	2.63V, 140ms
NS	ETC/MIC6315-26D4U	Open-Drain µP Reset Circuit	2.63V, 1100ms
NT	ETC/MIC8115TU	Microprocessor Reset Circuit	
NU	ETC/MIC6315-31D3U	Open-Drain µP Reset Circuit	3.08V, 140ms
NV	ETC/MIC8114TU	Microprocessor Reset Circuit	
NW	ETC/MIC6315-40D2U	Open-Drain µP Reset Circuit	
NX	ETC/MIC6315-46D3U	Open-Drain µP Reset Circuit	4.63V, 140 ms
NY	ETC/MIC6315-26D2U	Open-Drain µP Reset Circuit	2.63V, 20ms
NZ	ETC/MIC6315-46D4U	Open-Drain µP Reset Circuit	4.63V, 1100ms
PA	ETC/MIC6334-22D3U		
P30	MIC94030BM4	TinyFET® P-Channel MOSFET	
P31	MIC94031BM4	TinyFET® P-Channel MOSFET	with gate pull-up resistor
Rab	LM4040 b IM3- x.x LM4041 b IM3- x.x	Precision Micropower Shunt Voltage Reference x.x : output voltage (volts) a : voltage code b : tolerance code	
RBxx	MIC4043BM4	Low-Voltage Secondary-Side Shunt Regulator	
T10	MIC1555BM5	IttyBitty® Timer/Oscillator	
T11	MIC1557BM5	IttyBitty® Oscillator	
UDx	MIC2753	Power Supply Supervisor	active high
UEx	MIC2754	Power Supply Supervisor	active low
UFx	MIC2778	Voltage Monitor with Adjustable Hysteresis	

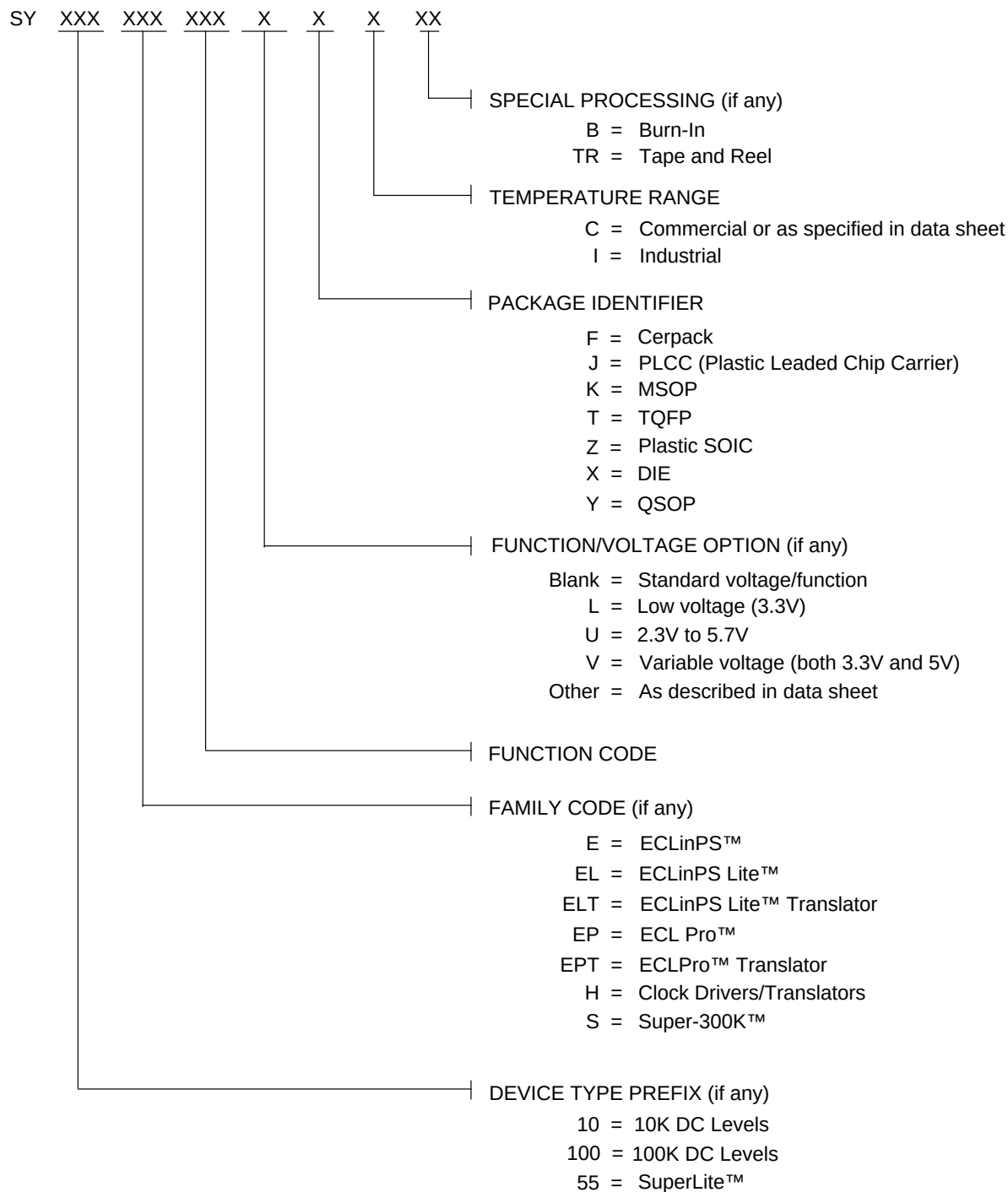
Communication Part Identification



ClockWorks™ Part Identification



Logic Part Identification



Worldwide Representatives and Distributors

Worldwide Sales Offices

Corporate Headquarters	2180 Fortune Dr.	San Jose, CA 95131	Tel: (408) 944-0800	Fax: (408) 944-0970
Eastern U.S. Area Sales Office	93 Branch St.	Medford, NJ 08055	Tel: (609) 654-0078	Fax: (609) 654-0989
Central U.S. Area Sales Office	8402 Sterling Suite 101	Irvine, TX 75063	Tel: (972) 929-0051	Fax: (972) 915-0120
Western U.S. Area Sales Office	3250 Scott Blvd.	Santa Clara, CA 95054	Tel: (408) 914-7670	Fax: (408) 914-7878
Northwest U.S. Regional Sales Office	P.o.b 152 401 N.e. Ravenna Blvd.	Seattle, WA 98115	Tel: (206) 526-7299	Fax: (206) 526-8829
Southwest U.S. Regional Sales Office	7545 Irvine Center Dr. Suite 200	Irvine, CA 92618	Tel: (949) 623-8433	Fax: (949) 623-8305
Southeast U.S. Regional Sales Office	8601 Six Forks Rd. Suite 400	Raleigh, NC 27609	Tel: (919) 676-5315	Fax: (919) 676-5316
Mexico, Central & South America	8402 Sterling Suite 101	Irving, TX 75063 USA	Tel: (972) 929-0051	Fax: (972) 915-0120
Micrel Semiconductor Asia Ltd.	4f. Jinsol Building 826-14, Yeoksam-dong	Kangnam-ku Seoul 135-080 Korea	Tel: + 82(2)3466-3000	Fax: + 82(2)3466-2999
Micrel Semiconductor, Taiwan	121-10, No. 237 Sec. 2, Fu-hsing South Rd.	Taipei, Taiwan, R.O.C.	Tel: + 886(2)2705-4976	Fax: + 886(2)2705-4977
Micrel Japan Sales Office	1-16-15, Dogenzaka, Shibuyaku	Tokyo 150-0043 Japan	Tel: + 81-3-5428-0871	Fax: + 81-3-5428-0872
Micrel Europe Technical Center	1st Floor, 3 Lockside Place, Mill Lane	Newbury, Berks RG 145QS UK	Tel: + 44(1635)524455	Fax: + 44(1635)524466

U.S. Sales Representatives

AL	CSR Electronics	Suite 931 303 Williams Ave.	Huntsville, AL 35801	Tel: (256) 533-2444	Fax: (256) 536-4031
AK	contact factory			Tel: (408) 944-0800	Fax: (408) 944-0970
AZ	Thom Luke Sales, Inc.	Suite A200 9700 North 91st St.	Scottsdale, AZ 85258	Tel: (480) 451-5400	Fax: (480) 451-0172
AR	Bravo Sales	Suite 375 16801 Addison Rd.	Dallas, TX 75248	Tel: (972) 250-2900	Fax: (972) 250-2905
CA	BAE Sales, Inc.	Suite 315W 2001 Gateway Pl.	San Jose, CA 95110	Tel: (408) 452-8133	Fax: (408) 452-8139
CA	CK Associates	Suite 102 8333 Clairmont Mesa Blvd.	San Diego, CA 92111	Tel: (858) 279-0420	Fax: (858) 279-7650
CA	Select Electronics	Suite 125 1651 E. Fourth St.	Santa Ana, CA 92701	Tel: (714) 648-0181	Fax: (714) 648-0669
CO	Lange Sales, Inc.	Suite 100, Building A 1500 West Canal Ct.	Littleton, CO 80120	Tel: (303) 795-3600	Fax: (303) 795-0373
CT	Datcom Technologies	One Evergreen Ave.	Hamden, CT 06518	Tel: (203) 288-7005	Fax: (203) 281-4233
DE	Midlantech Sales, Inc.	Suite E-2 607 Easton Rd.	Willow Grove, PA 19090	Tel: (215) 657-3370	Fax: (215) 657-3371
FL	Conley Associates	3696 Ulmerton Rd.	Clearwater, FL 33762	Tel: (727) 572-8895	Fax: (727) 572-8896
FL	Conley Associates	Suite 222 1750 West Broadway St.	Oviedo, FL 32765	Tel: (407) 365-3283	Fax: (407) 365-3272
GA	CSR Electronics, Inc.	Suite 120 3555 Koger Blvd.	Duluth, GA 30338	Tel: (678) 380-5080	Fax: (678) 380-5081
HI	contact factory			Tel: (408) 944-0800	Fax: (408) 944-0970
ID	Cascade Technical Sales	Suite 200 25 Central Way	Kirkland, WA 98033	Tel: (425) 822-7299	Fax: (425) 822-7479
ID	Lange Sales, Inc.	Suite 205 772 East 3300 South	Salt Lake City, UT 84106	Tel: (801) 487-0843	Fax: (801) 484-5408
IL	Sumer, Inc.	1675 Hicks Rd.	Rolling Meadows, IL 60008	Tel: (847) 991-8500	Fax: (847) 991-0474
IL	IRI of Kansas	Suite 200 101 Clark	St. Charles, MO 63301	Tel: (314) 298-8787	Fax: (314) 298-9843
IN	Technology Marketing Corp.	1526 East Greyhound Pass	Carmel, IN 46032	Tel: (317) 844-8462	Fax: (317) 573-5472
IN	Technology Marketing Corp.	4630-10 West Jefferson Blvd.	Fort Wayne, IN 46804	Tel: (219) 432-5553	Fax: (219) 432-5555
IN	Technology Marketing Corp.	1218 Appletree Ln.	Kokomo, IN 46902	Tel: (765) 454-2000	Fax: (765) 457-3822
IA	J.R. Sales Engineering	1930 St. Andrews, NE	Cedar Rapids, IA 52402	Tel: (319) 393-2232	Fax: (319) 393-0109
KS	IRI of Kansas	Suite 240 10000 College Blvd.	Overland Park, KS 66210	Tel: (913) 338-2400	Fax: (913) 338-0404
KY	Technology Marketing Corp.	Suite 1A 100 Trade St.	Lexington, KY 40511	Tel: (606) 253-1808	Fax: (606) 253-1662
LA	Bravo Sales	Suite 375 16801 Addison Rd.	Addison, TX 75248	Tel: (972) 250-2900	Fax: (972) 250-2905
ME	CTC Associates	12 Southwest Park	Westwood, MA 02090	Tel: (781) 320-1818	Fax: (781) 320-8282
MD	Tri-Mark, Inc.	1131L Benfield Blvd.	Millersville, MD 21108	Tel: (410) 729-7350	Fax: (410) 729-7364
MA	CTC Associates	12 Southwest Park	Westwood, MA 02090	Tel: (781) 320-1818	Fax: (781) 320-8282
MI	Technology Marketing Corp.	Suite 109 25882 Orchard Lake Rd.	Farmington Hills, MI 48336	Tel: (248) 473-8733	Fax: (248) 473-8840
MN	The Twist Company	3433 Broadway St., NE	Minneapolis, MN 55413	Tel: (612) 331-1212	Fax: (612) 331-8783
MI	CSR Electronics	Suite 931 303 Williams Ave.	Huntsville, AL 35801	Tel: (256) 533-2444	Fax: (256) 536-4031
MO	IRI of Kansas	Suite 200 101 Clark	St. Charles, MO 63301	Tel: (314) 298-8787	Fax: (314) 298-9843
MT	Lange Sales, Inc.	Suite 100, Building A 1500 West Canal Ct.	Littleton, CO 80120	Tel: (303) 795-3600	Fax: (303) 795-0373
NB	J.R. Sales Engineering	1930 St. Andrews, NE	Cedar Rapids, IA 52402	Tel: (319) 393-2232	Fax: (319) 393-0109
NV	BAE Sales, Inc.	Suite 315W 2001 Gateway Pl.	San Jose, CA 95110	Tel: (408) 452-8133	Fax: (408) 452-8139
NV	Thom Luke Sales, Inc.	Suite A200 9700 North 91st St.	Scottsdale, AZ 85258	Tel: (602) 451-5400	Fax: (602) 451-0172
NH	CTC Associates	12 Southwest Park	Westwood, MA 02090	Tel: (781) 320-1818	Fax: (781) 320-8282
NJ	Midlantech Sales, Inc.	Suite E-2 607 Easton Rd.	Willow Grove, PA 19090	Tel: (215) 657-3370	Fax: (215) 657-3371
NJ	Harwood Associates	355H River Rd.	Nutley, NJ 07110	Tel: (201) 865-4411	Fax: (973) 622-0565
NM	Thom Luke Sales, Inc.	Suite A200 9700 North 91st St.	Scottsdale, AZ 85258	Tel: (602) 451-5400	Fax: (602) 451-0172
NY	Harwood Associates	25 High St.	Huntington, NY 11743	Tel: (516) 673-1900	Fax: (516) 673-2848
NY	CTC Associates	12 Southwest Park	Westwood, MA 02090	Tel: (781) 320-1818	Fax: (781) 320-8282
NC	CSR Electronics, Inc.	Suite 2 5848 Faringdon Pl.	Raleigh, NC 27609	Tel: (919) 878-9200	Fax: (919) 878-9117
ND	The Twist Company	3433 Broadway St., NE	Minneapolis, MN 55413	Tel: (612) 331-1212	Fax: (612) 331-8783
OH	Technology Marketing Corp.	Suite 3 7775 Cooper Rd.	Cincinnati, OH 45242	Tel: (513) 984-6720	Fax: (513) 936-6515
OH	Technology Marketing Corp.	Suite 200 One Independence Pl., 4807 Rockside Rd.	Cleveland, OH 44131	Tel: (216) 520-0150	Fax: (216) 520-0190
OK	Bravo Sales	Suite 375 16801 Addison Rd.	Addison, TX 75248	Tel: (972) 250-2900	Fax: (972) 250-2905
OR	Cascade Technical Sales	Suite 244 15455 NW Greenbrier Pkwy.	Beaverton, OR 97006	Tel: (503) 645-9660	Fax: (503) 645-6363
PA	Midlantech Sales, Inc.	Suite E-2 607 Easton Rd.	Willow Grove, PA 19090	Tel: (215) 657-3370	Fax: (215) 657-3371
PA	Technology Marketing Corp.	Suite 200 One Independence Pl., 4807 Rockside Rd.	Cleveland, OH 44131	Tel: (216) 520-0150	Fax: (216) 520-0190
RI	CTC Associates	12 Southwest Park	Westwood, MA 02090	Tel: (781) 320-1818	Fax: (781) 320-8282
SC	CSR Electronics, Inc.	Suite 2 5848 Faringdon Pl.	Raleigh, NC 27609	Tel: (919) 878-9200	Fax: (919) 878-9117
SD	The Twist Company	3433 Broadway St., NE	Minneapolis, MN 55413	Tel: (612) 331-1212	Fax: (612) 331-8783
TN	CSR Electronics	Suite 931 303 Williams Ave.	Huntsville, AL 35801	Tel: (256) 533-2444	Fax: (256) 536-4031
TX	Bravo Sales	Suite 350 5910 Courtyard Dr.	Austin, TX 78731	Tel: (512) 328-7550	Fax: (512) 328-7426
TX	Bravo Sales	Suite 375 16801 Addison Rd.	Addison, TX 75248	Tel: (972) 250-2900	Fax: (972) 250-2905
TX	Bravo Sales	Suite 308 Willowbrook Pl. 1 17314 State Hwy. 249	Houston, TX 77064	Tel: (281) 955-7445	Fax: (281) 539-2728
TX	Thom Luke Sales, Inc.	Suite A200 9700 North 91st St.	Scottsdale, AZ 85258	Tel: (602) 451-5400	Fax: (602) 451-0172
UT	Lange Sales, Inc.	Suite 205 772 East 3300 South	Salt Lake City, UT 84106	Tel: (801) 487-0843	Fax: (801) 484-5408
VT	CTC Associates	12 Southwest Park	Westwood, MA 02090	Tel: (781) 320-1818	Fax: (781) 320-8282
VA	Tri-Mark, Inc.	1131L Benfield Blvd.	Millersville, MD 21108	Tel: (410) 729-7350	Fax: (410) 729-7364
WA	Cascade Technical Sales	Suite 200 25 Central Way	Kirkland, WA 98033	Tel: (425) 822-7299	Fax: (425) 822-7479
WA D.C.	Tri-Mark, Inc.	1131L Benfield Blvd.	Millersville, MD 21108	Tel: (410) 729-7350	Fax: (410) 729-7364
WV	Technology Marketing Corp.	Suite 200 One Independence Pl., 4807 Rockside Rd.	Cleveland, OH 44131	Tel: (216) 520-0150	Fax: (216) 520-0190
WI	The Twist Company	3433 Broadway St., NE	Minneapolis, MN 55413	Tel: (612) 331-1212	Fax: (612) 331-8783
WI	Sumer, Inc.	13555 Bishops Ct.	Brookfield, WI 53005	Tel: (262) 784-6641	Fax: (262) 784-1436
WY	Lange Sales, Inc.	Suite 100, Building A 1500 West Canal Ct.	Littleton, CO 80120	Tel: (303) 795-3600	Fax: (303) 795-0373

U.S. Distributors – Corporate Offices

Die Distribution

Chip Supply, Inc.	7725 Orange Blossom Trail	Orlando, FL 32810	Tel: (407) 298-7100	Fax: (407) 290-0164
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Packaged Devices

Newark Electronics	4801 North Ravenswood	Chicago, IL 60640	Tel: (773) 784-5100	Fax: (773) 907-5217
Arrow Electronics, Inc.	25 Hub Dr.	Melville, NY 11747	Tel: (800) 777-2776	Fax: (516) 391-4280
Nu Horizons Electronics Corp.	70 Maxess Rd.	Melville, NY 11747	Tel: (516) 396-5000	Fax: (516) 396-5050
Future Electronics	237 Hymus Blvd.	Pointe Claire, PQ H9R 5C7	Tel: (514) 694-7710	Fax: (514) 695-3707

International Sales Representatives and Distributors

Europe--Die Distribution ONLY

U.K.	Chip Supply, Inc.	5 Queen Street	Oldham OL1 1RD	Tel: + 44 (1616) 336627	Fax: + 44 (1616) 260380
U.K.	Eltek Semiconductor, Ltd.	Nelson Road Industrial Estate	Dartmouth Devon TQ6 9LA	Tel: + 44 (1803) 834455	Fax: + 44 (1803) 833011

International--Packaged Devices

Australia	BRAEMAC PTY LTD.	1/59-61 Burrow Rd.	Alexandria, Sydney 2015	Tel: 61 (2) 9550-6600	Fax: 61 (2) 9550-6377
Australia	Future Electronics	2nd Floor 1013 Whitehorse Rd. Box Hill Melbourne, Muehlgassee 86-88 Drehteststraat 7 1083 HK Amsterdam	Victoria 3128 A-2380 Perchtoldsdorf	Tel: 61 (3) 98997944 Tel: (43) 1 86 305-0	Fax: 61 (3) 98909632 Fax: (43) 1 86 305-98
Austria	CODICO GmbH & CoKG	Unit 1 2015 32nd Ave., NE	Netherlands	Tel: + 31 (20) 504 14 35	Fax: + 31 (20) 642 39 48
Belgium	Nijkerk Elektronika B.V.	10630 172 St.	A-2380 Perchtoldsdorf	Tel: (43) 1 86 305-0	Fax: (43) 1 86 305-98
Bulgaria	CODICO GmbH & CoKG	Suite 200 1144 29th Ave. NE	A-2380 Perchtoldsdorf	Tel: 359 (2) 560946	Fax: 359 (2) 560946
Bulgaria	Future Electronics	6029 103rd St.	1113 Sofia	Tel: + 55 (11) 231-0277	Fax: + 55 (11) 255-0511
Brazil	Artimar Ltda.	Suite 200 1144 29th Ave. NE	01223-000 São Paulo - SP	Tel: (403) 777-1550	Fax: (403) 777-1553
Canada	Enerlec Sales West ^(Rep.)	3671 Viking Way, #7	Calgary, AB T2H 2S8	Tel: (403) 236-4622	Fax: (403) 236-4366
Canada	Hewetson Company ^(Rep.)	8555 Commerce Ct.	Calgary, AB T2E 6Z3	Tel: (403) 291-5626	Fax: (403) 291-5444
Canada	Active Electronics	200-3689 East 1st. Avenue	Calgary, AB T2H 2M4	Tel: (403) 259-6817	Fax: (403) 259-8699
Canada	Arrow Electronics Inc.	200-3689 East 1st. Avenue	Edmonton, AB T5S 1H8	Tel: (403) 483-6266	Fax: (403) 484-8926
Canada	Arrow Electronics Inc.	1515 King Edward St.	Calgary, AB T2E 7P1	Tel: (780) 291-5333	Fax: (780) 291-5444
Canada	FAI	504-1780 Wellington Ave.	Edmonton, AB T6H 2H3	Tel: (780) 438-5888	Fax: (780) 436-1874
Canada	FAI	504-1780 Wellington Ave.	Calgary, AB T2E 7P1	Tel: (403) 291-3443	Fax: (403) 291-7054
Canada	Future Electronics	#213 95 Chernin Crowley Fourm	Richmond, BC V6V 1W1	Tel: (604) 273-0882	Fax: (604) 273-0884
Canada	Hewetson Company ^(Rep.)	161 Herzberg Rd.	Burnaby, BC V5A 4N4	Tel: (604) 421-2333	Fax: (604) 421-5030
Canada	Active Electronics	Unit 2 1350 Matheson Blvd.	Vancouver, BC V5M 1C2	Tel: (604) 654-1050	Fax: (604) 294-3170
Canada	Active Electronics	1023 Merivale Road	Vancouver, BC V5M 1C2	Tel: (604) 294-1166	Fax: (604) 294-1206
Canada	Arrow Electronics Inc.	88 Hines Rd.	Winnipeg, MB R3H 0R7	Tel: (204) 632-1260	Fax: (204) 633-6716
Canada	Arrow Electronics Inc.	171 Superior Blvd.	Winnipeg, MB R3H 1B2	Tel: (204) 786-3075	Fax: (204) 783-8133
Canada	Arrow Semiconductor	171 Superior Blvd.	Winnipeg, MB R3H 1B2	Tel: (204) 944-1446	Fax: (204) 783-8183
Canada	FAI	Suite 210 1101 Prince of Wales Dr.	Moncton, NB E1A 7R1	Tel: (506) 389-9991	Fax: (506) 389-9992
Canada	FAI	Suite 200 5935 Airport Rd.	Kanata, ON K2K 1X4	Tel: (613) 271-9044	Fax: (613) 271-9040
Canada	Future Electronics	Suite 210 1101 Prince of Wales Dr.	Mississauga, ON L4W 4M1	Tel: (905) 238-8825	Fax: (905) 238-2817
Canada	Newark Electronics	331 Consortium Ct.	Ottawa, ON K1Z 6A6	Tel: (613) 728-7900	Fax: (613) 728-3586
Canada	Newark Electronics	6200 Dixie Rd.	Kanata, ON K2K 2T8	Tel: (613) 271-8200	Fax: (613) 271-8203
Canada	Hewetson Company ^(Rep.)	Suite 597 100 Boul. Alexix Nihon	Mississauga, ON L5T 2L6	Tel: (905) 670-7769	Fax: (905) 670-7781
Canada	Active Electronics	Suite 190 1990 Boul. Charest Ouest	Mississauga, ON L5T 2L6	Tel: (905) 565-4405	Fax: (905) 565-4410
Canada	Arrow Electronics Inc.	Suite 100 1255 Trans Canada Hwy.	Ottawa, ON K2C 3W7	Tel: (613) 727-8622	Fax: (613) 727-2642
Canada	FAI	Ste. 201 1000 Ave. St. Jean Baptiste	Mississauga, ON L4V 1W5	Tel: (905) 820-8244	Fax: (905) 820-8046
Canada	FAI	1st Floor 1000 St. Charles Blvd.	Ottawa, ON K2C 3W7	Tel: (613) 727-1800	Fax: (613) 727-9819
Canada	Future Electronics	Ste. 201 1000 Ave. St. Jean Baptiste	London, ON N6E 2S8	Tel: (519) 685-4280	Fax: (519) 685-7104
Canada	Future Electronics	237 Hymus Blvd.	Mississauga, ON L5T 2E1	Tel: (905) 670-2888	Fax: (905) 670-1019
Canada	Newark Electronics	4480 Cote De Liesse	St. Laurent, PQ H4M 2P5	Tel: (514) 744-6511	Fax: (514) 744-0904
China	Galaxy Far East Corp.	Rm 0514 New Caohejing Tower	Ste. Foy, PQ G1N 4K8	Tel: (418) 682-5775	Fax: (418) 682-6282
		509 Caobao Road	Dorval, PQ H9P 2V4	Tel: (514) 421-7411	Fax: (514) 421-7430
China	Lestina International Ltd.	Rm 2403, Xiamen Commercial Bldg. No. 46,	Quebec, PQ G2E 5G5	Tel: (418) 877-1414	Fax: (418) 877-7070
		Zhichun Road Hai Dian District	Vaudreuil, QC J7V 8P5	Tel: (514) 457-3004	Fax: (514) 457-4912
China	Lestina International Ltd.	Rm 609, Yongfeng Bldg. No., 52	Quebec, PQ G2E 5G5	Tel: (418) 877-6666	Fax: (418) 877-6671
		Yongfeng Road High New Area,	Pointe Claire, PQ H9R 5C7	Tel: (514) 694-7710	Fax: (514) 695-3707
China	Lestina International Ltd.	Rm 3-3 Ke Chuang Bldg. 50, Yu Zhou Rd.	Mt. Royal, PQ H4N 2R1	Tel: (514) 738-4488	Fax: (514) 738-4606
		Goa Xin Ji Shu Kaifa Qu Shi Qiao Pu			
China	Lestina International Ltd.	Rm 03, 19/F Donghuan Tower No. 474	Shanghai 200233	Tel: + 86 (21) 64956484	Fax: + 86 (21) 64852237
		Donghuan Road			
China	Lestina International Ltd.	Rm 205 No. 29D Yudao Street	Beijing 100086	Tel: + 86 (10) 6257-6174	Fax: + 86 (10) 6257-6143
China	Lestina International Ltd.	Rm 806, Lawyer Hall,	Chengdu, Sichuan 610041	Tel: + 86 (28) 518-7012	Fax: + 86 (28) 518-7012
		1538 Zhong Shan West Rd.	Chongqing 400039	Tel: + 86 (23) 68619099	Fax: + 86 (23) 68608938
			Guangzhou 510075		Fax: + 86 (20) 87627227
			Nanjing 210016	Tel: + 86 (25) 4890188	Fax: + 86 (25) 4892089
			Shanghai 200233	Tel: + 86 (21) 6464-696	Fax: + 86 (21) 6464-4848

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Denmark	Future Electronics	Hoerskaetten 14, Conpenhagen	Taastrup 2630	Tel: + 45 43 55 00 20	Fax: + 45 43 55 00 25
Denmark	Future Electronics	Falrch-Huset	7500 Holstebro	Tel: + 45 96 10 09 61	Fax: + 45 96 10 09 62
Finland	Integrated Electronics Oy Ab	Laurinmäenkuja 3 A	00440 Helsinki	Tel: + 358 (9) 2535 4400	Fax: + 358 (9) 2535 4450
Finland	Future Electronics	Olarinluoma 7 Espoo	Helsinki 02200	Tel: + 358 9 525 9950	Fax: + 358 9 455 1050
France	CCI Electronique ^(Rep.)	12 Allee de la Vierge, Silic 577	94653 Rungis Cedex	Tel: + 33 (1) 41.80.70.00	Fax: + 33 (1) 46.75.32.07
France	Arrow Electronique, S.A.	73/79 Rue des Solets, Silic 585	94663 Rungis Cedex	Tel: + 33 (1) 49.78.49.00	Fax: + 33 (1) 41.80.96.30
France	FAI	Parc Technopolis -LP 854			
		3 Avenue du Canada Bat. Theta 2 Bat.	91974 Courtaboeuf Cedex	Tel: + 33 (1) 69.82.11.80	Fax: + 33 (1) 69.82.11.38
France	Future Electronics	Parc Technopolis -LP 854			
		3 Avenue du Canada Bat. Theta 2 Bat.	91974 Courtaboeuf Cedex	Tel: + 33 (1) 69.82.11.80	Fax: + 33 (1) 69.82.11.38
France	Future Electronics	Europarc du Chêne 4 Rue Edison Batiment A	69673 Lyon, Bron Cedex	Tel: + 33 (4) 72.15.86.00	Fax: + 33 (4) 72.15.00.68
France	Future Electronics	12, Allee des Peupliers Z.A. des Peupliers	35510 Cesson Seigne	Tel: + 33 (2) 23.45.60.80	Fax: + 33 (2) 23.45.60.79
France	Future Electronics	5 Avenue Albert Durand Aeropole 3	31700 Toulouse, Blagnac	Tel: + 33 (5) 62.74.72.40	Fax: + 33 (5) 62.74.14.72
Germany	ActiveComp GmbH ^(Rep.)	Mitterstrasse 12	85077 Manching	Tel: + 49 (8) 45 93 2140	Fax: + 49 (8) 45 93 21429
Germany	Dacom Sud	Freisinger Straße 13	85737 Ismaning	Tel: + 49 (89) 96 48 80	Fax: + 49 (89) 96 51 65
Germany	Future Electronics				
	Deutschland GmbH	München Straße 18	85774 Unterföhring	Tel: + 49 (89) 95 72 70	Fax: + 49 (89) 95 72 7173
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Germany	Retronic GmbH	Larchenstr 8	63549 Ronneburg	Tel: + 49 (61) 84 900590	Fax: + 49 (61) 84 900591
Germany	Retronic GmbH	Windweg 24	09116 Chemnitz	Tel: + 49 (371) 85 90 50	Fax: + 49 (371) 85 90 50
Germany	SASCO Semiconductor	Hermann-Oberth-Str. 16	85640 Putzbrunn	Tel: + 49 (89) 46 112330	Fax: + 49 (89) 46 112730
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Hungary	CODICO GmbH & CoKG	Muehlgassee 86-88	A-2380 Perchtoldsdorf	Tel: (43) 1 86 305-0	Fax: (43) 1 86 305-98
Hungary	Future Electronics	2nd Floor, Tartsay Vilmos UTCA 3	Budapest H-1126	Tel: + 36 (1) 224 0510	Fax: + 36 (1) 224 0511
India	Future Electronics	T8 & 9, No. 66 GEM Plaza, Infantry Rd.	Bangalore-1 560 001	Tel: + 91 (80) 558-7885	Fax: + 91 (80) 559-2995
India	Future Electronics	404, Gateway Plaza Hiranandari Gardens, Powai	Mumbai, Bombay 400 076	Tel: + 91 (22) 570-2745	Fax: + 91 (22) 570-2743
India	Future Electronics	1308-1310 Ansai Tower 38 Nehru Place	New Delhi 110 019	Tel: + 91 (11) 646-1414	Fax: + 91 (11) 646-1418
India	Hynetic Electronics	78/1, 3rd Cross Gavipuram Extension	Bangalore 560 019	Tel: + 91 (80) 652-0852	Fax: + 91 (80) 652-0473
India	Hynetic Electronics	E-245, Amar Colony Lajpat Nagar 4	New Delhi 110 024	Tel: + 91 (11) 628-2434	Fax: + 91 (11) 628-2435
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India	Hynetic Electronics	Flat No. C103 1st Floor, J.B. Nagar Ajit Nagar Owners Assoc.	Andheri East Mumbai 400 059	Tel: + 91 (22) 822-3575	Fax: + 91 (22) 822-3575
Ireland	Focus Electronics Distribution Ltd.	Cottis House Locks Hill Bradley Way	Rochford Essex SS4 1BB	Tel: + 44 1702 542301	Fax: + 44 1702 542302
Ireland	Future Electronics Ltd.	Future House Poyle Road Colnbrook	Berkshire SL3 0E2	Tel: + 44 (1753) 763000	Fax: + 44 (1753) 689100
Ireland	Silicon Concepts Ltd.	PEC Lynchborough Road Passfield, Lipphook	Hampshire GU30 7SB	Tel: + 44 (1428) 751617	Fax: + 44 (1428) 751603
Ireland	Solid State Supplies ^(Stocking Rep.)	Eastlands Lane Paddock Wood	Kent TN12 6BU	Tel: + 44 (1892) 836836	Fax: + 44 (1892) 837837
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Israel	Future Electronics	103 Medinat Hayehudim Street POB 2219	46120 Herzeliya	Tel: + 972 (9) 970 1414	Fax: + 972 (9) 958 4333
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Italy	Future Electronics	Via Domenico Turazza 30	Padova 31528	Tel: + 39 (49) 899 20111	Fax: + 39 (49) 807 0582
Japan	Hakuto Co. Ltd. ^(Rep.)	Nagoya-Seni Bldg. 9-27, Nishiki, 2-chome Naka-ku,	Nagoya Aichi 460	Tel: + 81 (52) 204-8910	Fax: + 81 (52) 204-8935
Japan	Hakuto Co. Ltd. ^(Rep.)	292-4, Asouda-machi Matsuyama	Ehime 790	Tel: + 81 (89) 931-8910	Fax: + 81 (89) 945-6218
Japan	Hakuto Co. Ltd. ^(Rep.)	Felix Iwai Bldg. 2-3,			
		Hakataekiminami, 3-chome Hakata-ku	Fukoka 812	Tel: + 81 (92) 431-5330	Fax: + 81 (89) 431-5265
Japan	Hakuto Co. Ltd. ^(Rep.)	3-18, Miyanomae, 2-chome, Itami	Hoygo 664	Tel: + 81 (72) 784-8910	Fax: + 81 (72) 784-7860
Japan	Hakuto Co. Ltd. ^(Rep.)	56, Takehanatokenokaido-cho Yamashina-ku	Kyoto 607	Tel: + 81 (75) 593-8910	Fax: + 81 (75) 593-8990
Japan	Hakuto Co. Ltd. ^(Rep.)	Kamisugikokune Bldg.			
		4-10, Kamisugi, 1-chome, Aoba-kuSendai,	Miyagi 980	Tel: + 81 (22) 224-8910	Fax: + 81 (22) 224-0645
Japan	Hakuto Co. Ltd. ^(Rep.)	1-13, Shinjuku, 1-chome, Shinjuku-ku	Tokyo, 160-8910	Tel: + 81 (3) 3225-8910	Fax: + 81 (3) 3225-9001
Japan	Nippon Imex Corporation	No. 6 Sanjo Bldg., 5F 1-46-9 Matsubara			
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Japan	Shinden Hightex	Esu Bldg. 1-6-1 Higashiyama, Meguro-ku,	Tokyo 156	Tel: + 81 (3) 3719-8585	Fax: + 81 (3) 3719-8668
Korea	GenTech Corporation ^(Stocking Rep.)	6F, Daewon Bldg. 67-5, Yangjae-dong			
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Korea	UTO International ^(Rep.)	2F, 835-61, Yeoksam-dong Kangnam-ku	Seoul 135-081	Tel: + 82 (2) 501-5665	Fax: + 82 (2) 508-3250
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Romania	CODICO GmbH & CoKG	Muehlgassee 86-88	A-2380 Perchtoldsdorf	Tel: (43) 1 86 305-0	Fax: (43) 1 86 305-98
Scotland	Future Electronics	Mirran Court (3) 123 Renfrew Road	Paisley PA4 4EA	Tel: + 44 (141) 840 6500	Fax: + 44 (141) 849 6971
Singapore	JAG Components (Pte.) Ltd.	Ruby Industrial Complex			
		Genting Block 80 Genting Lane, #11-06A	Singapore 349565	Tel: + 65 749 56 63	Fax: + 65 749 56 62
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Slovakia	CODICO GmbH & CoKG	Muehlgassee 86-88	A-2380 Perchtoldsdorf	Tel: (43) 1 86 305-0	Fax: (43) 1 86 305-98
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Spain	Future Electronics	Avenida del Parenon 10	Madrid 28042	Tel: + 34 (91) 721 4270	Fax: + 34 (91) 721 1043
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Sweden	Quantum Electronics AB ^(Rep.)	Girovägen 13	175 62 Jarfalla	Tel: + 46 (8) 506 498 70	Fax: + 46 (8) 506 498 99
Sweden	Future Electronics	Energigatan 10B	Kungsbacka 434 37	Tel: + 46 300 759 95	Fax: + 46 300 759 96
Sweden	Future Electronics	c/o Regus Business Centre Amiralsgatan 20	Malmo SE-211 55	Tel: + 46 (40) 664 35 51	Fax: + 46 (40) 664 35 41
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Taiwan, R.O.C.	Galaxy Far East Corp.	1F, No. 15 Alley 20 Lane.544			
		Sec. 1, Kuang Fu Road	Hsinchu	Tel: + 886 (3) 578-6766	Fax: + 886 (3) 577-4795

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